SAM VOLUME I

Quantum routes, shared fields and retained physical records

Chapter VI

Abstract. Coherent alternatives, physical correlation and a readable record are distinct objects. We integrate the original Chapter VI with its completed continuum, source-decoder and detector/memory supplements. The full original source, signed currents, even energy, two field polarizations, both counterterms, hard dressing, soft input and angular complements are retained. A new driven receiving construction places two prescribed moving source arms in one continuum Fock field. Its moving-frame generator includes field momentum, admits common-form dynamics, and gives an energy-based error relative to the isolated proper-age reference. A closed-arm experiment with a finite, paid coherent source encoding joins that dynamics to the actual local detector and retained record. New exact six-lane arithmetic gives ready error below 0.0016 under the stated budgets. This encoded example is separate from an unforced gravitational age signal. Multi-history coherence, repeated-read error, all outcomes, latency and work are carried explicitly. Born probabilities, unitarity and instruments remain receiving premises; microscopic routing/actuator calibration and a distant intrinsic source port remain named physical assignments.

Reading map. The chapter follows phase and correlation, readable records, restoration, the original source clock, its source-only decoder, the actual detector, and the new shared-field/coherent join. The prior supplement statements are integrated as mathematical sections, not restarted campaigns. Their historical provenance and original receiving assumptions remain explicit. New constructions and finite parameter choices in this chapter are developed in campaigns voli_chvi_foundation1, voli_chvi_source_decoder1, voli_chvi_detector_record1 and voli_chvi_shared_routes1 (17 October 2026; H002700, H002701, H002706), with finite matrices checked by native SLC-GEN5-R1 calls. Provenance trace: Courtroom II case voli_provenance/01.

1. A route can carry phase before it carries a record

A detector need not be watched for it to change an experiment. A physical interaction can correlate a route with another degree of freedom, and that correlation can change the interference seen when the route is recombined. The further question is whether the correlation remains readable, by which apparatus, and for how long. These are physical questions about a carrier and its dynamics, not about an observer's awareness.

Unified Volume I calls the transition to a durable accessible record a substrate write.[1] Its quantum section retains the Born rule and action phase, while deferring the detailed write algebra to the computational volumes. The present chapter expands that interface. It does not infer an amplitude evolution from a route label or turn an accessibility convention into a new collapse law.

Chapters II, III and VII provide an explicit interacting source, its clock current, and a source-to-field interface. Chapter IV separates local clock evolution from signal transport. Chapter V distinguishes a traveler's history from what a receiver can obtain and from what a prescribed boundary allows.[2, 3, 4, 5, 6] The quantum continuation asks what happens when alternatives remain coherent through part of that chain.

Three questions guide the calculation:

1. What phase and amplitude does a declared dynamics assign to each coherent alternative?
2. Which physical degrees of freedom become correlated with those alternatives, and how does that change interference?
3. What record can a specified domain actually decode, retain and transmit?

A free photon is not unwritten because it is massless. It can carry energy, momentum and phase without yet leaving the kind of persistent record being discussed. Nor does an inaccessible correlation stop affecting a reduced quantum state. Accessibility concerns which records a domain can use; the full joint evolution still contains correlations outside that domain.

For an orthonormal set of exclusive route-readout alternatives, normalized amplitudes give

$$P(j) = \frac{|a_j|^2}{\sum_k |a_k|^2}. \tag{1}$$

This is the retained Born premise.[11] Coherent alternatives reaching the same complete final outcome contribute to a common amplitude before its squared magnitude is taken. Arbitrary decompositions of one state do not define additional mutually exclusive outcomes. After a record forms, subsequent and conditional measurements still obey the same probability rule.

2. What the recovered source record establishes

The recovered CR075 package reports a scoped structural ledger-write pass. Its runner inspects source text, manifests, disclosures and named bridge entries. The retained QP reports supply constructed route tables and threshold selectors. Their presence is evidence of a specified computational interpretation, not a derivation of arbitrary complex amplitudes or a physical detector dynamics.[7]

This distinction matters for the probability account. QP014 normalizes constructed capacity, coherence-survival and access weights. It does not establish that these weights equal $|a_j|^2$ for arbitrary quantum preparations. QP001, QP010 and QP016 likewise do not establish universal physical decoherence or outcome thresholds. This chapter therefore uses their route/write

distinction while deriving its interference calculations from explicitly stated quantum laws.

The later exact-write sources supply reversible finite address updates and separately retained history data. A signed update followed by its inverse can restore an address while a retained event list remains different. That is a useful exact algebraic example of restoration without deletion. It is not yet a microscopic measurement interaction.[8]

The lifecycle source also keeps an important boundary open: the maps from its computational `A_OPERATOR` to contact and to `X1` are not supplied as a completed physical law. That typed operator must not be silently identified with the scalar accumulation coordinate $A(r)$ or with a Hamiltonian coupling.[9] The task here is to state what such a bridge would need to reproduce.

All recovered repository bytes are pinned and authenticated in the companion package. The complete CR075 directory was inspected; its original 242-input upstream environment was not recovered or rerun. Historical reported control flags are not recast as newly executed perturbed quantum experiments. The derivations and checks below are a new independent receiving calculation, not an upgrade of those historical receipts.

3. Three phases and one consistent action account

3.1 A signed Write integral is not a complete history

The original lifted source phases define

$$\begin{aligned} \gamma_{\mathrm{W}} &= \phi_a + \phi_b - 2\phi_p, \\ \chi &= \phi_p, \end{aligned} \tag{2}$$

with the legal-path one-form

$$\omega_{\mathrm{Write}} = \hbar(\mathrm{d}\gamma_{\mathrm{W}} + 3\mathrm{d}\chi). \tag{3}$$

Consequently its integrated signed action is

$$S_{\mathrm{Write}} = \hbar(\Delta\gamma_{\mathrm{W}} + 3\Delta\chi). \tag{4}$$

The lifted endpoints, wrapped phases, ordered event word and elapsed duration preserve different information. Chapter VII's retained histories can have equal endpoint and equal signed action while retaining different positive durations and event orders.[2] An exact differential does not reconstruct an ordered word from its integral.

The interacting phase $\theta = -\arg\langle Z\rangle$ is a separate observable coordinate. An interferometer's relative phase is another operand, obtained from its full controlled amplitude. No equality among γ_{W}, θ and the observed fringe phase follows simply because all three are called phases. Their identification requires an explicit coupling and readout.

3.2 An explicit massive-route comparison

For a declared test-body action, the massive point contribution is

$$S_{\mathrm{mass},j} = -m_0c^2 \int_{\gamma_j} \mathrm{d}\tau. \tag{5}$$

Here m_0c^2 must be a part of the energy not already counted in the internal generator. In the weak static comparison with $F = 1 - A$ to first order, $A \ll 1$, and $v^2/c^2 \ll 1$,

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = 1 - \frac{A}{2} - \frac{v^2}{2c^2} + O(A^2, Av^2/c^2, v^4/c^4). \tag{6}$$

For two routes with the same coordinate duration, the common rest phase cancels and

$$\Delta\varphi_{\mathrm{mass}} = \frac{m_0}{2\hbar} \int \left[c^2(A_1 - A_0) + (v_1^2 - v_0^2)\right] \mathrm{d}t + \text{higher orders}. \tag{7}$$

This is the ordinary phase of kinetic energy minus Newtonian potential energy with $\Phi = -c^2A/2$. It is conditional on the stated action and geometry, not a derivation of an A-dependent quantum law from the source grammar.

An interferometer also needs the phases of its preparation, splitting, routing and recombination operations, with recoil and every record retained by those controls. A comparison of two held clocks is not automatically a coherent recombination experiment. For static segments of duration T, the proper-age difference is $T(\sqrt{F_1} - \sqrt{F_0})$ under the declared metric; the apparatus must still make both alternatives and their interference available.

Equation (5) is not a photon phase prescription: a null route has zero proper time. A photon requires its wave equation and boundary-value or eikonal phase, with the source and receiver controls included. The earlier photon-road delay does not by itself define a quantum-state amplitude.

4. Quantum routes and physical records

Evidence status. The calculations below are consequences of standard finite-dimensional quantum mechanics. Born probabilities, Hilbert-space composition, and quantum instruments are premises. The SAM contribution at this stage is a proposed interpretation and a specification of what a substrate write must implement, not a new probability law or an experimentally distinct prediction.

The route language of Unified Volume I, §5, separates an unresolved alternative from a physical record. Its useful content becomes testable only after the record carrier, decoder, accessible domain, and retention interval are declared. A route can carry phase without leaving a readable mark. Conversely, correlations can suppress local interference even when no observer can read their remote carrier. Accessibility therefore qualifies the *record available to a domain*; it does not switch ordinary quantum entanglement or decoherence on and off.

4.1 A controlled marker and its observable fringe

Let S be a two-route system with orthogonal states $|0\rangle, |1\rangle$, and let M be a marker initially in $|m\rangle$. Write $q = 1 - p$, $0 \le p \le 1$, and prepare

$$|\psi\rangle_S = \sqrt{p}\,|0\rangle + e^{i\phi}\sqrt{q}\,|1\rangle. \tag{8}$$

A path-preserving marker coupling has the form

$$\begin{aligned} U &= |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1, \\ |m_j\rangle &= U_j|m\rangle. \end{aligned} \tag{9}$$

The joint state and overlap convention are

$$|\Psi\rangle = \sqrt{p}\,|0\rangle|m_0\rangle + e^{i\phi}\sqrt{q}\,|1\rangle|m_1\rangle, \tag{10}$$
$$\gamma = \langle m_1|m_0\rangle. \tag{11}$$

Tracing out M, rather than assuming a collapse, gives

$$\rho_S = \begin{pmatrix} p & \sqrt{pq}\,e^{-i\phi}\gamma \\ \sqrt{pq}\,e^{i\phi}\gamma^* & q \end{pmatrix}. \tag{12}$$

For a balanced output analyser with projectors onto $|\pm_\theta\rangle = (|0\rangle \pm e^{i\theta}|1\rangle)/\sqrt{2}$, the Born rule yields

$$P_\pm(\theta) = \tfrac{1}{2} \pm \sqrt{pq}\,\mathrm{Re}[e^{i(\theta-\phi)}\gamma]. \tag{13}$$

Thus the scanned single-port visibility is

$$V = \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}} = 2\sqrt{pq}\,|\gamma|. \tag{14}$$

These expressions assume a lossless two-route preparation, a balanced analyser, a phase scan, and no outcome-dependent filtering. Unequal route populations already limit V without a marker. A global phase in the overlap shifts the fringe; its magnitude controls the contrast. A change in contrast is consequently not, by itself, proof of a durable record.

The action-phase interface remains the ordinary one: for two coherent routes, $\Delta\phi = (S_1 - S_0)/\hbar$. An A-dependent action must specify its Lagrangian, integration parameter, and units before it predicts a fringe shift. The schematic symbol L_A in the source text supplies no independent microscopic Hamiltonian and does not derive the Born rule. In particular, probabilities continue to govern subsequent measurements after a record has formed; "before the write" describes the proposed accounting language, not a restriction on the applicability of quantum probability.

4.2 Distinguishability, with the priors retained

Consider the operational task of guessing a route from its marker, given priors p, q. For any binary POVM E_0, E_1 on M,

$$P_{\text{guess}} = p\,\mathrm{Tr}(E_0\rho_0) + q\,\mathrm{Tr}(E_1\rho_1), \qquad \rho_j = |m_j\rangle\langle m_j|. \tag{15}$$

Set $H = p\rho_0 - q\rho_1$. Since $E_1 = I - E_0$, maximizing $q + \mathrm{Tr}(E_0 H)$ selects the positive spectral subspace of H. Hence

$$P_{\text{guess}}^{\text{opt}} = \tfrac{1}{2}(1 + D), \qquad D = \|p\rho_0 - q\rho_1\|_1. \tag{16}$$

This is weighted Helstrom distinguishability,[15] not the unweighted trace distance unless $p = q = 1/2$. On the span of the two pure markers, $\mathrm{Tr}\,H = p - q$ and $\det H = -pq(1 - |\gamma|^2)$. The two eigenvalues have opposite signs or one is zero, so

$$D = \sqrt{1 - 4pq|\gamma|^2}, \qquad D^2 + V^2 = 1. \tag{17}$$

Identical marker states give $D = |p - q|$: success then comes only from the prior imbalance. Orthogonal markers give $D = 1$, $V = 0$. Equation (17) is an equality for this pure joint state with full marker access and the specified analyser. It is not an unrestricted equality for every apparatus. It belongs to the established two-way duality framework; see Englert.[12]

There is a short general proof of the relevant inequality. Let ρ_{SM} be any joint state and purify it with E. For $0 < p < 1$ the purification can be written

$$|\Omega\rangle = \sqrt{p}\,|0\rangle|u\rangle_{ME} + \sqrt{q}\,|1\rangle|v\rangle_{ME}, \tag{18}$$

with normalized conditional vectors. If $c = \langle 0|\rho_S|1\rangle$, then $c = \sqrt{pq}\langle v|u\rangle$ and the same analyser gives $V = 2|c|$. Weighted pure-state discrimination on ME gives $D_{ME} = \sqrt{1 - 4|c|^2}$. Partial trace is a trace-norm contraction on Hermitian operators, yielding

$$D_M \le D_{ME}, \qquad D_M^2 + V^2 \le 1. \tag{19}$$

The endpoints $p = 0, 1$ follow directly. This proof also covers a restricted accessible subsystem of the marker. It does not license comparing visibility in one postselected ensemble with distinguishability in another. These are the standard channel and discrimination conventions.[10]

4.3 Reversal, erasure, and inaccessible markers

An exact reversible two-qubit fixture uses $|m\rangle = |0\rangle$, $U_0 = I$, and

$$U_1 = R(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}. \tag{20}$$

Then $\gamma = \cos\alpha$. At $p = 1/2$, $V = |\cos\alpha|$ and $D = |\sin\alpha|$. At $\alpha = \pi/2$ the marker is orthogonal and the reduced route state has no coherence. Applying the *joint controlled inverse* $U^\dagger$ before analysis restores $|\psi\rangle|0\rangle$ and the original visibility. This is a reversal of the correlation-generating interaction. A marker-only unitary cannot accomplish it, since it preserves $\langle m_1|m_0\rangle$.

A quantum eraser is a different operation. For the orthogonal marker, measure M in $|\pm\rangle_M = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Each outcome has probability $1/2$ and prepares the conditional route state

$$|\psi_\pm\rangle = \sqrt{p}\,|0\rangle \pm e^{i\phi}\sqrt{q}\,|1\rangle. \tag{21}$$

Its fringe visibility is $2\sqrt{pq}$, but the two fringes are shifted by π. Averaging their distributions cancels the interference. Conditional coincidence sorting requires the marker outcome; it does not restore an unconditional remote fringe. Outcome-dependent feed-forward can align the phases, but uses a communicated outcome and an operation on S. This distinction is central to the original quantum-eraser proposal of Scully and Drühl and to conditional coincidence experiments.[13, 14]

Environmental copies constrain reversal. For branch states of the factorized form $|m_j\rangle|e_j\rangle$, the coherence multiplier is

$$\gamma_{ME} = \langle m_1|m_0\rangle\langle e_1|e_0\rangle. \tag{22}$$

Reversing the controlled S–M coupling while leaving E intact does not remove the environmental factor. Orthogonal $|e_j\rangle$ maintain zero local coherence even when E is inaccessible. Generic environmental states need not factorize; the purification

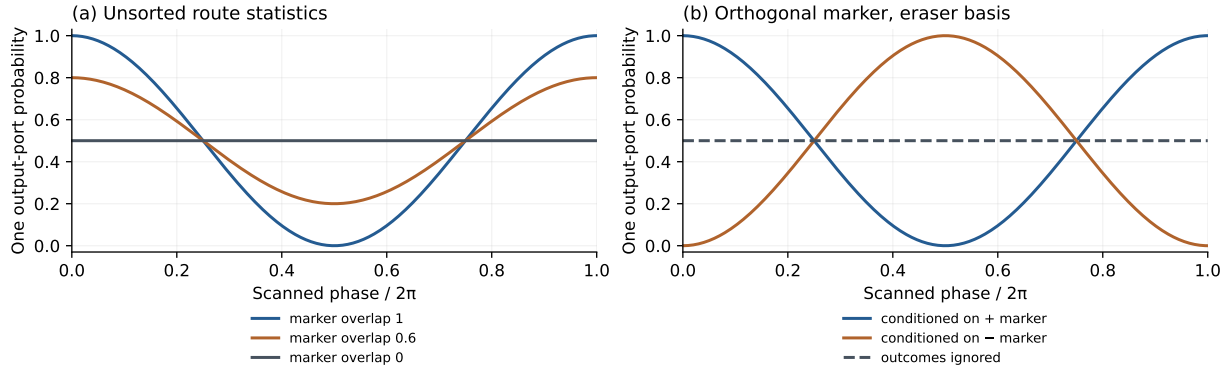


Figure 1: Exact balanced two-route comparator. (a) Real marker overlap changes unsorted interference through $P_+ = (1+\gamma\cos\varphi)/2$. (b) For orthogonal markers, measuring the complementary marker basis gives two conditional fringes, each with probability one half. Ignoring that outcome yields the unchanged flat marginal. These analytic finite-system curves illustrate the stated protocol; they are not measured SAM data or an original-field simulation.

argument above replaces this product formula in that case. No assertion about a gravitational horizon follows merely from tracing out E.

More generally, let Λ_M be any trace-preserving local channel, including an unread measurement or reset. For every bipartite state,

$$\rho_S' = \mathrm{Tr}_M[(I\otimes\Lambda_M)(\rho_{SM})] = \rho_S, \tag{23}$$

$$T(\rho_S',\rho_S) = \tfrac{1}{2}\|\rho_S'-\rho_S\|_1 = 0. \tag{24}$$

The equality follows from $\Lambda_M^*(I) = I$: every local expectation on S is unchanged. Thus no route-only POVM can reveal the remote channel choice. Selective outcomes can change a conditional state, but the selection information must be transmitted. This is the precise no-signalling control on any proposed accessibility or erasure narrative.

4.4 A durable accessible record is an operational claim

A useful record definition specifies more than entanglement. Let $\{\mathscr{I}_j\}$ be a quantum instrument: each map is completely positive and trace non-increasing, and $\sum_j \mathscr{I}_j$ is trace preserving. Its probabilities are $p_j = \mathrm{Tr}\,\mathscr{I}_j(\rho)$, and its conditional output is $\mathscr{I}_j(\rho)/p_j$ for $p_j > 0$. The instrument can describe a declared path-marking task, but must not be read as establishing a pre-existing route value in every coherent preparation. If only selected outcomes are retained, the omitted outcomes must remain in the complete instrument; renormalizing the retained branch describes postselection, not a new unconditional probability law.

For a proposed memory R, declare (i) the operational labels j, (ii) the conditional memory states σ_R^j, (iii) the storage dynamics $\mathscr{N}_t$, (iv) the channel $\mathscr{A}_{\mathscr{D},t}$ delivering data to domain $\mathscr{D}$, and (v) an allowed decoder POVM $\{F_j(t)\}$. Call it an (ε,T) readable record for that domain when a predeclared physically implementable decoder satisfies

$$\inf_{0\le t\le T}\ \min_{j:p_j>0}\ \mathrm{Tr}\Big[F_j(t)\mathscr{A}_{\mathscr{D},t}\mathscr{N}_t(\sigma_R^j)\Big] \ge 1-\varepsilon. \tag{25}$$

Here $0 \le \varepsilon < 1/2$ for a binary record. The minimum over labels prevents a strongly biased source from passing by always guessing the common label. The time origin is the end of writing, and the interval tests one read performed at any chosen time; repeated non-destructive reading requires an additional back-action specification. Fixed decoders are a valid special case. Access may incorporate transport delay, controllable operations, or a restricted measurement algebra; the protocol must state which. Without those declarations, "accessible" is not a quantitative physical property.

Four different statements must remain separate. Entanglement concerns the joint state. Decoherence concerns suppressed off-diagonal terms in a specified reduced state and basis. A single registered outcome is described operationally by an instrument and the Born rule. A persistent readable record additionally passes Eq. (25). Unitary correlation and partial trace alone do not select one unique global outcome or solve the measurement problem.[16] Nor does a record require awareness: every element of the definition is a physical preparation, channel, or decoder.

4.5 Controls and the boundary of the claim

The preceding equations prescribe concrete wrong-controls. Treating a marker-only rotation as a joint inverse incorrectly predicts an unconditional restored fringe. Tracing out an orthogonal environmental marker and then declaring it absent incorrectly reinstates coherence. Sorting eraser outcomes and comparing that conditional visibility with unsorted which-way data mixes ensembles. Treating D as marker-acquired information without subtracting or reporting prior predictability misreads biased preparations. Treating a returned local state as a globally erased event discards the history carrier. Finally, labelling a reduced density matrix as one selected outcome imports an additional postulate without saying so.

These controls are standard quantum benchmarks. They discipline the route/write terminology and identify what an executable substrate account must reproduce. No new SAM prediction follows from renaming the instrument outcome a write. Such a prediction would require an additional, quantitatively specified dynamics or access rule, its parameter domain, and an experiment separating it from the retained quantum

premises.

5. Readable records and retained route coherence

The following statements use the quantum state, channel, and measurement premises already declared in Chapter VI. They make the relation between a readable record and an unsorted interference experiment quantitative. The record labels must refer to the same routes and the same ensemble as the visibility; a reliable record of an unrelated label supplies no such bound.

5.1 A same-state, worst-label theorem

Let S be the two-route system and let M contain the remaining physical marker degrees of freedom. At the chosen observation time write an arbitrary normalized joint state as

$$\begin{aligned} \rho_{SM} &= \sum_{j,k=0}^{1} |j\rangle\langle k| \otimes R_{jk}, \\ p = \mathrm{Tr} R_{00}, &\quad q = \mathrm{Tr} R_{11} = 1-p. \end{aligned} \tag{26}$$

For $p,q>0$, the conditional marker states are $\sigma_M^0 = R_{00}/p$ and $\sigma_M^1 = R_{11}/q$. Their labels specify the two diagonal route alternatives used by this operational test; this notation does not select a unique outcome in a coherent joint state.

Declare a label-independent channel $\Lambda_t : M \longrightarrow D$ that includes the storage and access operations available to the reading domain at time t. It is completely positive and trace preserving, and acts on the marker rather than on S. Let $F_0(t), F_1(t)$ be one physically allowed decoder POVM on D, with $F_0 + F_1 = I_D$. The decoder may depend on the declared time and controls, but receives no copy of the unknown label. Define

$$\begin{aligned} \tau_D^j(t) &= \Lambda_t(\sigma_M^j), \\ r_j(t) &= \mathrm{Tr}[F_j(t)\tau_D^j(t)]. \end{aligned} \tag{27}$$

Storage and access can instead be specified jointly on an enlarged marker containing a correlated environment, a transport carrier, and retained classical records. No product-state or independent-noise assumption is required. An environment correlated with the marker must be included in that joint channel description when the proposed reduced channel would otherwise depend on its correlations. Conditional states for the theorem are always the actual conditional states of Eq. (26) under that description.

Let $c = \langle 0|\rho_S|1\rangle$. For a lossless two-route preparation and the declared balanced, variable-phase analyser, with no outcome-dependent filtering, the unsorted single-port visibility is

$$V = 2|c|. \tag{28}$$

The local, unselected channel Λ_t does not change ρ_S. Consequently a decoder can be analysed on D while the visibility refers to this same route marginal. A joint controlled inverse or feed-forward operation on S would define a different experiment and must be included before applying the theorem to its resulting state.

Theorem. If, at the chosen time, both conditional decoder successes satisfy

$$\begin{aligned} \min\{r_0(t), r_1(t)\} &\geq 1-\varepsilon, \\ 0 \leq \varepsilon &< \tfrac{1}{2}, \end{aligned} \tag{29}$$

then

$$\boxed{V \leq 4\sqrt{pq\,\varepsilon(1-\varepsilon)} \leq 2\sqrt{\varepsilon(1-\varepsilon)}.} \tag{30}$$

For balanced priors the two bounds agree. For unequal priors the first retains their additional restriction on the fringe. At $p=0$ or $q=0$, positivity already gives $c=0$ and $V=0$; a conditional success statement about an absent label must be supplied by a separately declared calibration preparation if it is wanted.

Derivation. Apply the decoder to $(I_S \otimes \Lambda_t)(\rho_{SM})$ and retain its two outcomes. The unnormalized route state for outcome y is positive, with diagonal entries pa_y, qb_y, where

$$\begin{aligned} a_y &= \mathrm{Tr}[F_y \tau_D^0], \\ b_y &= \mathrm{Tr}[F_y \tau_D^1]. \end{aligned} \tag{31}$$

If its off-diagonal entry is c_y, positivity of this 2×2 matrix and preservation of the unselected marginal give

$$\begin{aligned} |c_y| &\leq \sqrt{pq\,a_y b_y}, \\ c &= \sum_y c_y, \\ |c| &\leq \sqrt{pq}\sum_y \sqrt{a_y b_y}. \end{aligned} \tag{32}$$

This argument retains arbitrary off-diagonal marker blocks and arbitrary correlations with inaccessible degrees of freedom.

Put $a = a_1$ and $b = b_0$, the two conditional error probabilities. The readability hypothesis gives $0 \leq a, b \leq \varepsilon$, and the last sum in Eq. (32) is

$$\sqrt{(1-a)b} + \sqrt{a(1-b)}. \tag{33}$$

Write $a = \sin^2\alpha$, $b = \sin^2\beta$, with $0 \leq \alpha, \beta \leq \arcsin\sqrt{\varepsilon} < \pi/4$. This sum is $\sin(\alpha+\beta)$, hence it is at most $2\sqrt{\varepsilon(1-\varepsilon)}$. Multiplying by $2\sqrt{pq}$ gives Eq. (30), including $\varepsilon = 0$ by continuity or direct substitution.

The weaker bound also follows immediately from weighted discrimination: the stated decoder has prior-weighted success at least $1-\varepsilon$, so $D_D = \|p\tau_D^0 - q\tau_D^1\|_1 \geq 1-2\varepsilon$. The same-state inequality $D_D^2 + V^2 \leq 1$ then gives $V \leq 2\sqrt{\varepsilon(1-\varepsilon)}$. The Bhattacharyya derivation uses both separate error constraints and therefore retains the stronger prior-dependent factor. No optimal decoder needs to be physically available: the actual allowed decoder in Eq. (29) is enough.

Tight qubit example. For any $0 \leq \varepsilon < 1/2$, take

$$|m_0\rangle = \sqrt{1-\varepsilon}|0\rangle + \sqrt{\varepsilon}|1\rangle, \tag{34}$$

$$|m_1\rangle = \sqrt{\varepsilon}|0\rangle + \sqrt{1-\varepsilon}|1\rangle, \tag{35}$$

$$|\Psi\rangle = \sqrt{p}\,|0\rangle|m_0\rangle + e^{i\phi}\sqrt{q}\,|1\rangle|m_1\rangle. \tag{36}$$

The computational-basis decoder succeeds with probability $1-\varepsilon$ on each label, and $\langle m_1|m_0\rangle = 2\sqrt{\varepsilon(1-\varepsilon)}$. Its visibility saturates the first bound in Eq. (30) for every p,q. For $\varepsilon = 0$ the marker is perfectly readable and the route marginal has zero visibility, while the pure joint state still retains its coherent superposition. A later joint reversal can recover interference while changing the record. The theorem bounds the specified unsorted fringe; it does not identify loss of its contrast with deletion of global coherence.

5.2 Arrival, retention, and the time of reading

An (ε, W_D) readable record for domain D requires the worst-label condition at every allowed reading time $t \in W_D$, with the channel, allowed decoder, and time origin declared. The theorem then bounds the contemporaneous visibility at every such time for which the same route/marker description applies. The statement concerns one read at a chosen time. A sequence of reads requires the actual reading instruments and their disturbance; the interval condition alone supplies no nondestructive repeated-read guarantee.

There is a precise causal pre-arrival restriction. Suppose a newly encoded label has no label-dependent receiver state in its causal past and the declared channel has not delivered any label-dependent degree of freedom by time t. Then

$$\tau_D^0(t) = \tau_D^1(t) = \tau_D(t). \tag{37}$$

Every binary decoder obeys $r_0 + r_1 = 1$ on this common state, so

$$\min_j r_j(t) \le \tfrac{1}{2}. \tag{38}$$

Thus it cannot satisfy the declared $\varepsilon < 1/2$ readability before arrival. This conclusion needs the explicit identical-state premise, provided physically by the causal/no-signalling access model. A receiver that already holds a correlated copy of the label is a different resource configuration. Unequal priors can improve an average guess before arrival, but cannot pass the worst-label condition.

The storage window and receiver window are consequently different objects. Let the end of writing be $t = 0$, and suppose the stored encoding is guaranteed at the memory for emission times $t_e \in [0, T_M]$. For a fixed latency τ and a separately budgeted readout time τ_{dec}, the corresponding idealized completion times are

$$\begin{aligned} t_D &= t_e + \tau + \tau_{\mathrm{dec}}, \\ W_D &= [\tau + \tau_{\mathrm{dec}},\, T_M + \tau + \tau_{\mathrm{dec}}]. \end{aligned} \tag{39}$$

This translated window assumes that the access channel actually samples the memory at t_e and that its transport, intervening storage, and decoder errors have been paid. A protocol needing the memory throughout a finite read instead requires that entire memory-side interval inside its retention window. Variable travel times use the declared retarded emission map $t_e(t_D)$ and its admissible interval. Starting a receiver guarantee at the end of writing would incorrectly include times before delivery unless delivery was already available. Choosing the domain time origin at arrival is valid when that convention is stated.

5.3 Composable error budgets and retained failed preparations

Worst-label budgets compose without independent errors. One convenient quantum formulation includes all classical outcome flags and relevant environmental references in the joint state. Let the ideal decoder have error at most ε_0 on each label. Suppose a hybrid comparison of the actual preparation, storage, transport, and readout gives trace-distance errors δ_k on the relevant joint conditional states at stage k. Each stage bound must hold for the hybrid inputs actually used, including retained correlations; an entangled reference is included in the comparison when present. Equivalently, a uniform channel bound can be supplied on that declared joint input class.

Contractivity under the subsequent trace-preserving channels and the triangle inequality give a final joint-state distance at most $\sum_k \delta_k$. A fixed POVM outcome probability changes by no more than that distance. Consequently

$$\begin{aligned} r_j^{\mathrm{actual}}(t) &\ge 1 - \varepsilon_0 - \sum_k \delta_k, \\ \varepsilon_{\mathrm{tot}} &= \varepsilon_0 + \sum_k \delta_k. \end{aligned} \tag{40}$$

The readability and visibility theorem applies when the declared total is below $1/2$. Isolated marginal error estimates cannot be substituted for these joint-state bounds when unaccounted correlations change the channel. The hybrid channels and flags also preserve any feedback used by the actual instrument.

A classical alternative is an explicit sequence of bad-event flags whose conditional probability, on every reached prior history and each label, is at most δ_k. If the final decoder error on the all-good branch is at most ε_0, the union bound gives the same additive budget. This argument likewise needs no independence. It does need the stated conditional bounds; adding unrelated average failure rates would not establish a worst-label guarantee.

Preparation failure belongs in the full instrument. Suppose label j has failure probability $f_j \le f$, and the successful preparation has worst-label reading error at most $\varepsilon_{\mathrm{succ}}$. If an explicit failure or abstention is counted as an unsuccessful attempted read, then

$$r_j^{\mathrm{attempt}} \ge (1 - f_j)(1 - \varepsilon_{\mathrm{succ}}) \ge 1 - f - \varepsilon_{\mathrm{succ}}. \tag{41}$$

Success, failure, and abstention flags remain in the normalized all-outcome state and in the physical resource account. A larger decoder alphabet can be retained explicitly, or its failure outcomes can be assigned a stated binary guess without discarding them.

Alternatively a claim may be conditioned on a heralded success. Its normalization, conditional record states, and conditional visibility must then all use that same accepted ensemble. In a route-preserving heralded preparation the accepted priors become

$$\begin{aligned} p_{\mathrm{acc}} &= \frac{p(1-f_0)}{p(1-f_0) + q(1-f_1)}, \\ q_{\mathrm{acc}} &= 1 - p_{\mathrm{acc}}, \end{aligned} \tag{42}$$

when the denominator is positive. Outcome-dependent filtering can also change the coherence, so the accepted joint state itself

must be used. A conditional high-visibility ensemble cannot be compared with a readable record certified only in a different, unselected ensemble. No reset of failed preparations or deletion of their records is supplied by this probability accounting.

5.4 A finite readable-label capacity consequence

For N declared labels, suppose the full accessible record input has finite Hilbert dimension d and one POVM $\{F_j\}_{j=1}^N$ satisfies $\mathrm{Tr}(F_j\sigma_j) \geq 1-\varepsilon$ for every normalized conditional state σ_j. Since $0 \leq \sigma_j \leq I$,

$$N(1-\varepsilon) \leq \sum_{j=1}^{N} \mathrm{Tr}(F_j\sigma_j) \leq \sum_{j=1}^{N} \mathrm{Tr}\, F_j = d. \tag{43}$$

Thus $d \geq \lceil N(1-\varepsilon)\rceil$. Any correlated side carrier available to the decoder belongs in that accessible dimension; alternatively the decoder is pulled back to a POVM on the declared input space. This bound concerns N physically decodable labels. Merely naming additional ordered histories does not add distinguishable record states. It assigns neither an energy per label nor a thermal erasure cost.

6. Restoration of a local state and retention of a physical history

The source interface is

$$W8 + X1 \longrightarrow W9 \longrightarrow W8'. \tag{44}$$

Here $W8$ names the resting, non-contact local substrate state; $X1$ names the supplied interaction; and $W9$ names the temporary write state. The prime marks return after interaction, not a new substrate atom. The source's intended distinction is that a local state can return while an event remains recorded elsewhere. This section gives that distinction an explicit quantum-information comparator. It does not identify the symbols $W8$, $X1$, and $W9$ with Hilbert-space dimensions, derive their exact SAM/SLC grammar, or prove a substrate microdynamics.

Chapter II already distinguishes an address from the history of its use. Returning to a retained address does not by itself say whether a record survives. Nor does a finite alphabet supply unlimited storage: a physical history requires distinguishable states of some specified recipient. The same discipline applies here. We must name the restored subsystem, the other degrees of freedom, and the operations by which a record can be read.

6.1 What a unitary reset can and cannot do

Let L be a local record cell and let E include every other degree of freedom participating in a closed finite-dimensional evolution. Suppose $\{|m_i\rangle_L\}_{i=1}^n$ are orthonormal pointer states and E starts in a fixed pure state $|e_0\rangle$. An exact reset of L has the form

$$U\big(|m_i\rangle_L|e_0\rangle_E\big) = |0\rangle_L|e_i\rangle_E. \tag{45}$$

Reset–recipient proposition. If U is unitary, then $\langle e_i|e_j\rangle = \delta_{ij}$. Consequently E cannot also return to one fixed state when $n > 1$. The recipient needs at least n mutually orthogonal states.

Proof. Preservation of inner products gives

$$\delta_{ij} = \langle m_i, e_0|m_j, e_0\rangle = \langle 0, e_i|0, e_j\rangle = \langle e_i|e_j\rangle. \tag{46}$$

An exact pure local output factors from the rest, so this argument also covers a purported reset whose output was initially written as a general joint pure state. A mixed initial environment may be purified by adding its reference to E; the conclusion then concerns that enlarged recipient. It does not guarantee a record readable from an arbitrarily chosen proper part of E. □

There is an important scope condition. If a separate system S already contains a correlated copy, the input is not the fixed-environment input of Eq. (45). In the more general pure-state mapping

$$|s_i\rangle|m_i\rangle|e_0\rangle \longmapsto |s_i'\rangle|0\rangle|e_i\rangle, \tag{47}$$

unitarity requires, for $i \neq j$,

$$\langle s_i'|s_j'\rangle\langle e_i|e_j\rangle = 0. \tag{48}$$

If S returns to one common pure state, the recipient states must be orthogonal. If S instead retains orthogonal alternatives, it can itself carry the distinction; a new record in E is not compulsory. For example, reversing a controlled write resets the cell while retaining the source bit. The assertion that "every reset must write a new history register" would therefore be false. The invariant is distinguishability of the complete closed state, not the permanence of a selected local mark. For mixed or entangled subsystems, surviving information may be in joint correlations rather than readable in any one marginal.

6.2 Three registers: a write, a restoration, and a retained marker

Use three qubits: route S, local cell L, and history recipient H. The history register is initially blank. Define

$$\begin{aligned} C|r,\ell,h\rangle &= |r,\ell\oplus r,h\rangle,\\ T &= \mathrm{SWAP}_{LH}, \end{aligned} \tag{49}$$

where $r,\ell,h \in \{0,1\}$ and $\oplus$ is addition modulo two. Both maps permute the full eight-state computational basis, so both are unitary without a restriction to the intended input states. Their composition is

$$\begin{aligned} U &= TC,\\ U|r,\ell,h\rangle &= |r,h,\ell\oplus r\rangle. \end{aligned} \tag{50}$$

For a normalized coherent route input,

$$\begin{aligned} |\Psi_{\mathrm{in}}\rangle &= (\alpha|0\rangle+\beta|1\rangle)_S|0\rangle_L|0\rangle_H,\\ |\Psi_{\mathrm{write}}\rangle &= \alpha|000\rangle+\beta|110\rangle, \end{aligned} \tag{51}$$

$$\begin{aligned} |\Psi_{\mathrm{restored}}\rangle &= \alpha|000\rangle+\beta|101\rangle\\ &= |0\rangle_L\big(\alpha|00\rangle_{SH}+\beta|11\rangle_{SH}\big). \end{aligned} \tag{52}$$

The local cell is exactly restored. Conditioned on the route alternative, the history states are $|0\rangle_H$ and $|1\rangle_H$, which can be perfectly discriminated. For a coherent input, however, the final state is an entangled superposition, not a derivation of one selected outcome. The comparator exhibits a readable pointer correlation and local restoration. It does not derive collapse or a unique realized history.

The analogy to Eq. (44) is deliberately limited: resting cell, interaction-enabled write, restored cell with a retained recipient correlation. The interaction $X1$ is not treated as an extra qubit

or an arithmetical increment. The temporary $W9$ designation may refer to the interaction configuration; it need not mean that the cell alone has changed in every branch. In particular, the $r = 0$ branch leaves its pointer value at zero even during the controlled write.

The inverse exists on every input: $U^{-1} = CT$. With joint control of S, L, H, it returns Eq. (52) to $|\Psi_{\text{in}}\rangle$. The phase information was never deleted from the complete state. By contrast, tracing out the record gives

$$\rho_S = |\alpha|^2 |0\rangle\langle 0| + |\beta|^2 |1\rangle\langle 1| \tag{53}$$

both after the write and after the restoration. Thus the swap does not restore route-only interference. Global coherence is preserved throughout; its observability depends on access to the correlated degrees of freedom. Joint reversal, or suitable measurements of the record with conditioned route statistics, can reveal that coherence. An operation on the inaccessible record alone cannot change the unconditioned route marginal.

For example, take $\alpha = 1/\sqrt{2}$ and $\beta = e^{i\phi}/\sqrt{2}$. The joint observable $X_S X_H$ has expectation $\cos\phi$ after restoration, whereas every route-only expectation is independent of ϕ. This is the precise distinction between preservation in the closed system and accessibility in a restricted domain.

This device stores one binary marker. It does not store an arbitrary sequence of writes. If H is already occupied, the swap returns that previous content to L, rather than resetting L to zero. Repeated restoration needs additional blank capacity, a deliberate reversible compression when possible, or an explicitly modeled disposal channel. Perfectly distinguishing N possible complete histories requires N orthogonal global record states and hence a record-space dimension at least N. This is a storage statement, not a heat formula.

6.3 Pointer copying is not universal quantum cloning

For a declared orthogonal pointer basis, a controlled operation can copy its label:

$$|r\rangle|0\rangle \longmapsto |r\rangle|r\rangle. \tag{54}$$

Applied to a superposition, it creates the correlated state in Eq. (51); it does not produce two copies of $\alpha|0\rangle + \beta|1\rangle$. Indeed, let the overlap of two normalized inputs be z and allow the cloner to finish in input-dependent auxiliary states $|a_\psi\rangle$ and $|a_\varphi\rangle$. Inner-product preservation would require $z = z^2\langle a_\psi|a_\varphi\rangle$. For $0 < |z| < 1$, this implies $1 \le |z|$, a contradiction. Universal cloning is therefore impossible even when the auxiliary state is allowed to change.

Transfer is different. A swap implements

$$|\psi\rangle_L|0\rangle_H \longmapsto |0\rangle_L|\psi\rangle_H \tag{55}$$

for every $|\psi\rangle$, preserving its entanglement with any reference. The input remains present once, at its new location. Resetting L while leaving both the recipient and all other degrees of freedom fixed would instead delete an unknown input, contrary to inner-product preservation. These elementary statements suffice for this interface; no universal copying or unexplained destruction of quantum information is being assumed. Related no-deleting and no-hiding results have their own hypotheses; no claim that information must reside only in correlations is used here.[19, 18]

6.4 A physical record needs an accessible decoder

The decoder criterion of Eq. (25) applies to the nominated recipient, not merely the restored cell. For equal priors, the three-register example has trace distance one between its conditional history states and zero between its conditional local-cell states. The history permits perfect route decoding; the restored cell permits only chance performance. The Helstrom optimum is attainable only if its measurement belongs to the allowed set.

Durability adds a time and noise specification. A useful requirement is that a stated decoder recover the label with error at most ε after a declared storage interval and channel. Orthogonality at the moment of writing does not by itself establish durability. For example, encoding a classical bit in an odd number n of identical pointer cells and reading by majority vote gives

$$P_{\text{err}} \le \exp\{-2n(\tfrac{1}{2} - p)^2\}, \qquad p < \tfrac{1}{2}, \tag{56}$$

if, conditional on the stored bit, the cell errors are independent and each has probability at most p. For error indicators E_j and $t = 1/2 - p > 0$, independence and the bounded-variable exponential-moment inequality give

$$\Pr\left\{ \sum_j (E_j - \mathbb{E}E_j) \ge nt \right\} \le \inf_{\lambda > 0} e^{-\lambda nt + n\lambda^2/8} = e^{-2nt^2}. \tag{57}$$

Majority failure is contained in the displayed event. Correlated failures need not satisfy it: one common flip of all cells leaves a majority decoder with the same error probability as a single cell. This example is an optional classical storage mechanism, not a derived SAM durability law.

Finally, a history description is not an additional quantum observable. If two proposed histories yield the same accessible density operator, every accessible POVM gives the same outcome probabilities. Calling one state $W8$ and the other $W8'$ cannot change this. If a history can be read, its evidence must reside in distinguishable accessible states or correlations, with a specified decoding operation. This is the physical content of the prime: local return is compatible with a retained record, but the notation alone neither guarantees that record nor makes it accessible.

No thermodynamic cost is inferred from these unitary maps. A Landauer bound belongs to a separately declared cyclic erasure protocol with a thermal reservoir, temperature, initial logical ensemble, and work/heat accounting. Neither the number of local addresses nor the logarithm of a record-space dimension is, by itself, dissipated heat. The present result is narrower: a local restoration can transfer a physical distinction; it cannot establish the disappearance of that distinction from a closed unitary account.

7. The original interacting clock on coherent routes

7.1 The full interacting source as a route marker

A route-dependent clock requires a receiving rule connecting the local interacting dynamics to spacetime transport. The following rule is conditional [17]: the same isolated, co-moving source–field Hamiltonian acts along each prescribed timelike route, with its Chapter VII time chart replaced by accumulated proper time. External beam splitters, recombination, routing forces, and the spacetime geometry are controls; they are not derived from the source Hamiltonian. Tidal changes of the internal Hamiltonian, radiation caused by the controls, and inaccessible environmental records require additional terms or a larger marker system. A field shared nonlocally by both arms does not automatically admit this isolated co-moving reduction.

Retain the complete source space $\mathbb{C}^4 \otimes \mathbb{C}^4$ and the two polarizations of the continuum Fock field. In the Chapter VII dimensionless chart the rest generator is

$$\begin{aligned} H_0 &= \varepsilon(S + Q_L + Q_R + Q_L Q_R) + H_f \\ &\quad - \varepsilon(D\phi_+ + C\phi_\times) \\ &\quad + \varepsilon^2 \mu (D^2 + C^2), \\ D &= Q_L - Q_R, \qquad C = iSD, \qquad \mu = (4R)^{-1}. \end{aligned} \tag{58}$$

This is the original interacting form-sum Hamiltonian, not a bare two-level substitution. Its six two-state lanes and four unforced diagonal states are all retained. In a chosen prepared lane $a < b$, $d = q_a - q_b < 0$, $g = \varepsilon d$, and the restriction is $H_0 = \varepsilon e I + \varepsilon S + H_f - g(\sigma_z \phi_+ + \sigma_y \phi_\times) + cI$, where $e = q_a + q_b + q_a q_b$ and $c = 2g^2\mu$. Both the even energy and the two counterterms remain in the generator.

For route j, define

$$\begin{aligned} \eta_j &= s_0 \int_{\gamma_j} d\tau_{\text{proper}}, \qquad U_j = e^{-iH_0\eta_j}, \\ U_{\text{ctrl}} &= \sum_{j=1}^{2} |j\rangle\langle j| \otimes U_j. \end{aligned} \tag{59}$$

Here $s_0 = 1/t_u$ in the declared Chapter VII chart; no new SI calibration is inferred. With $I_* = \hbar$, the physical rest energy is $(\hbar/t_u)H_0$. Conditional on this transport law, the actual clock coordinate obeys $d\theta/dt = \nu(\eta_j)(d\tau_{\text{proper}}/dt)/t_u$ wherever its certified inverse exists. This does not extend the Chapter VII finite-window rate certificate beyond its stated preparation and interval.

Let the route input be $\sqrt{p_1}|1\rangle + e^{i\alpha}\sqrt{p_2}|2\rangle$, $p_1 + p_2 = 1$, initially uncorrelated with a marker state ρ. Tracing the marker gives route coherence proportional to

$$\begin{aligned} \Gamma(\delta\eta) &= \mathrm{Tr}(\rho U_2^\dagger U_1) = \mathrm{Tr}(\rho e^{-iH_0\delta\eta}), \\ \delta\eta &= \eta_1 - \eta_2. \end{aligned} \tag{60}$$

An ideal variable-phase recombiner measures fringe visibility $\mathscr{V} = 2\sqrt{p_1 p_2}|\Gamma|$. The standard Born measurement rule is assumed in this calculation, not derived from the original Write histories. The phase of Γ shifts the fringes; its modulus controls their contrast. Extra uncontrolled route records multiply or otherwise modify this coherence and must be included explicitly.

For balanced priors, the operational which-route distinguishability in the full marker is

$$\begin{aligned} \mathscr{D} &= \tfrac{1}{2}\|U_1 \rho U_1^\dagger - U_2 \rho U_2^\dagger\|_1, \\ \mathscr{D}^2 + |\Gamma|^2 &\le 1. \end{aligned} \tag{61}$$

Equality holds for a pure initial full marker with unrestricted joint measurements. For a mixed marker the inequality follows by purifying ρ: the two purified states have overlap Γ, and partial trace contracts trace distance. Restricted access to the source alone can only decrease $\mathscr{D}$. For unequal priors the corresponding statement uses $\|p_1\rho_1 - p_2\rho_2\|_1$ and $\mathscr{V}$. In particular, if $[\rho, H_0] = 0$, then $\rho_1 = \rho_2$ and $\mathscr{D} = 0$, although $|\Gamma|$ may be small. Mixed-state phase averaging is not itself accessible which-route information in that marker. Nor does the phase-clock expectation $z = \langle Z \rangle$ determine Γ: one is an expectation of an evolved observable, the other a full-state propagator overlap.

7.2 The needed spectral operand and its domain

Let P_{H_0} be the spectral measure of the full interacting Hamiltonian. The new original-source operand needed for exact route contrast is

$$\begin{aligned} d\mathscr{P}_\rho(E) &= \mathrm{Tr}[\rho P_{H_0}(dE)], \\ \Gamma(s) &= \int e^{-iEs}\, d\mathscr{P}_\rho(E). \end{aligned} \tag{62}$$

Chapter VII's bounds on M, Ω, z, and ν do not provide this entire measure or the exact Loschmidt amplitude. A new spectral enclosure, a controlled evaluation of this matrix element, or an overlap enclosure with a stated error is needed for a numerical full-continuum fringe curve. The matched hard mean energy $\bar{E}$ is also required for its absolute reference phase; a gap enclosure alone does not determine that phase.

If $\int E^2 d\mathscr{P}_\rho(E) < \infty$, set $\sigma_H^2 = \int (E - \bar{E})^2 d\mathscr{P}_\rho(E)$. Then

$$\begin{aligned} |\Gamma(s)|^2 &= 1 - \sigma_H^2 s^2 + o(s^2), \\ |\Gamma(s)| &= 1 - \tfrac{1}{2}\sigma_H^2 s^2 + o(s^2). \end{aligned} \tag{63}$$

These coefficients use the full prepared joint-energy distribution. Neither the bare exchange gap 2ε nor the calibration expectation $E_{\text{ref}} = 573049$ is an energy variance. The latter belongs to a different placement-basis calibration preparation. For a pure state the requisite condition is $\Psi \in D(H_0)$, not merely the energy form domain $D(H_f^{1/2})$. Chapter VII explicitly has ultraviolet-divergent $\int_0^\infty |u_R|^2 d\omega$; consequently a generic bare vacuum argument cannot supply this second moment.

When this finite-variance condition holds, the standard quantum speed bound [20, 21] follows because the projective Hilbert-space speed of a purification evolving under H_0 is σ_H. Thus

$$\begin{aligned} \arccos|\Gamma(s)| &\le \sigma_H|s|, \\ |\Gamma(s)| &\ge \cos(\sigma_H|s|), \quad \sigma_H|s| \le \pi/2. \end{aligned} \tag{64}$$

Any zero of the amplitude requires $|s| \ge \pi/(2\sigma_H)$. For a mixed initial marker this is a bound on the specified amplitude (or its purification), not a claim that the reduced marker becomes distinguishable at that time. The full pure-state marker also obeys $\mathscr{D} \le \sin(\sigma_H|s|)$ on this interval.

A finite full-H second moment for the actual declared preparation. The ultraviolet qualification does not invalidate the particular hard-branch preparation that Chapter VII actually declares. Let H_{ref} be its matched hard-interacting Hamiltonian, with the same even scalar and soft counterterm as H_0, and let

$$\begin{aligned} \psi_k &= \Phi_k \otimes |0_s\rangle, & \Psi_0 &= (\psi_0+\psi_1)/\sqrt{2}, \\ H_{\mathrm{ref}}\psi_k &= E_k\psi_k, & E_1-E_0 &= \Omega. \end{aligned} \tag{65}$$

The normalized exact hard eigenvectors are operator-domain vectors of their hard Hamiltonian. The reference preparation is therefore in $D(H_{\mathrm{ref}})$. Write

$$\begin{aligned} V_s &= H_0 - H_{\mathrm{ref}} \\ &= -g(\sigma_z\phi_{+,s} + \sigma_y\phi_{\times,s}), \\ v_s &= \int_0^\lambda |u_R(\omega)|^2 \mathrm{d}\omega \le v_s^{\mathrm{up}} = \frac{R^4\lambda^6}{675\kappa\pi}. \end{aligned} \tag{66}$$

On this soft vacuum, $V_s\Psi_0$ is a Hilbert-space vector with exactly one soft photon. The two polarization insertions are orthogonal and both Pauli operators preserve norm, giving $\|V_s\Psi_0\|^2 = 2g^2 v_s$. The form identity against every test vector in the common form domain is represented by the Hilbert vector $H_{\mathrm{ref}}\Psi_0 + V_s\Psi_0$. The representation theorem for the closed form therefore proves $\Psi_0 \in D(H_0)$ and $H_0\Psi_0 = H_{\mathrm{ref}}\Psi_0 + V_s\Psi_0$. Orthogonality of zero and one soft photon sectors now gives the new exact identity

$$\begin{aligned} \bar{E} &= \tfrac{1}{2}(E_0+E_1), \\ \boxed{\sigma_H^2 = \frac{\Omega^2}{4} + 2g^2 v_s} &\le \frac{\Omega^2}{4} + 2g^2 v_s^{\mathrm{up}}. \end{aligned} \tag{67}$$

This is a full interacting variance, despite its expression in matched reference data. It relies on the exact hard eigenbranch preparation; it is not a general domain assertion about arbitrary coherent source states. It also does not assert $\Psi_0 \in D(H_0^2)$ or existence of higher moments. The signed g is retained in the Hamiltonian; its square appears here because this is a norm. A rigorous numerical speed enclosure follows by using the original certificate's upper bounds for Ω, $|g|$, R, and λ, with their declared dependencies, rather than rounded lane numbers.

Six-lane consequences of the retained inputs. Table 1 uses the original rational enclosures for Ω, signed g, v_s^{up} and $T_0 = 2\pi/\Delta$. The new rational operations are recorded in the companion check output; each printed bound is verified outward by exact rational comparison. These are independently derived enclosures, not a new native-certificate run. The lower orthogonality ratio is slightly above one half in every lane, while the reference-amplitude error at T_0 is at most 4.54×10^{-11}. This does not imply that an exact zero or a perfect revival occurs.

An independent overlap enclosure from the same preparation. The reference overlap is $\Gamma_{\mathrm{ref}}(s) = e^{-i\bar{E}s}\cos(\Omega s/2)$. The reference orbit is a finite sum of operator-domain eigenvectors, and its soft insertion is Hilbert-space continuous. Strong

Lane	$10^6\sigma_H$ upper	error at T_0 upper	$s_\perp/T_0$ lower
01	1.745051466079	1.759×10^{-12}	0.500000001207
02	1.745051452191	3.644×10^{-12}	0.500000005186
03	1.745048665211	4.536×10^{-11}	0.500000803726
12	1.745051465446	1.886×10^{-12}	0.500000001389
13	1.745048878445	4.360×10^{-11}	0.500000742629
23	1.745049097788	4.172×10^{-11}	0.500000679781

Table 1: Full interacting hard-branch preparation: variance-speed and matched-reference bounds. The error is an amplitude error, and $s_\perp$ is any first possible overlap zero, in the dimensionless proper-time chart.

Hilbert-space Duhamel, with the full interacting propagator left intact, gives

$$\begin{aligned} &\|(e^{-isH_0} - e^{-isH_{\mathrm{ref}}})\Psi_0\| \\ &\quad\le \int_0^{|s|} \|V_s e^{-iuH_{\mathrm{ref}}}\Psi_0\| \mathrm{d}u = |s|\sqrt{2g^2 v_s}. \end{aligned} \tag{68}$$

The same orthogonal soft-sector norm is constant for every reference time; negative s is handled by reversing the integral. Consequently

$$\begin{aligned} \varepsilon_s &= |s|\sqrt{2g^2 v_s^{\mathrm{up}}}, \\ |\Gamma(s) - e^{-i\bar{E}s}\cos(\Omega s/2)| &\le \varepsilon_s, \\ |\Gamma(s)| &\ge \max\{0, |\cos(\Omega s/2)| - \varepsilon_s\}. \end{aligned} \tag{69}$$

This is a newly derived overlap estimate using Chapter VII's exact preparation, not an inference from its phase-rate certificate. It preserves all photon sectors in the actual propagation. The bound can become uninformative at long times; it does not certify phase-clock invertibility there.

For another preparation lacking a verified second moment, a safe alternative is spectral truncation. For a bounded interval I, let $p = \mathrm{Tr}[\rho P_{H_0}(I)] > 0$ and normalize its restricted spectral measure. Its amplitude Γ_I satisfies $|\Gamma - p\Gamma_I| \le 1-p$ and hence $|\Gamma| \ge \max\{0, p|\Gamma_I| - (1-p)\}$. A finite-variance speed bound for the restricted measure may then be used, but the tail probability $1-p$ must itself be established for the original state. A finite matrix cutoff without that tail is a comparator, not a continuum certificate.

7.3 Action phases, energy offsets, and route records

The coherent amplitude of a controlled route may contain a scalar factor $e^{iS_{\mathrm{ext},j}/\hbar}U_j$. Its scalar interference phase is $(S_{\mathrm{ext},1} - S_{\mathrm{ext},2})/\hbar$, combined with the phase of Γ. This accounting must partition the generator once. If an assigned rest mass m_0 is excluded from H_0, its point-particle contribution is $S_{\mathrm{mass},j} = -m_0c^2\int_{\gamma_j} d\tau_{\mathrm{proper}}$. If that same energy is already included in the internal generator, adding this phase again double counts it. The same warning applies to an even source-energy offset and the scalar counterterm in a fixed lane.

Explicitly, replacing H_0 by $H_0 + bI$ changes $\Gamma(s)$ to $e^{-ibs}\Gamma(s)$ without changing visibility, the same-history expectation z, or which-route distinguishability. Thus an energy offset invisible to the original clock-rate equation can produce a relative route phase $-b(\eta_1 - \eta_2)$. An unchanged total amplitude requires the compensating assignment $S_{\mathrm{ext},j} \mapsto S_{\mathrm{ext},j} + \hbar b\eta_j$. A lane-dependent even operator is not a common scalar on a superposition of lanes and can affect its full spectral distribution. The

original signed Write integral, the quantum rest-generator phase, and a relativistic point-particle action are therefore distinct operands until a receiving identification and a non-overlapping energy account are supplied.

Finally, distinct ordered route or Write records need not imply distinct accessible quantum states. If two histories induce the same observable state and leave no coupled record, an extra history label alone changes no measurement probability. If a physical record is produced, it belongs in the marker or environment and its overlap controls the residual coherence. This allows different histories with equal endpoint and equal signed action without silently creating an additional observable or a new clock law.

Independent finite-source comparator. The companion executable `check_clock_routes.py` retains the exact 16-state source matrices, signed D, C, the full even source energy, both polarizations, and both discrete counterterms. It checks source calibration, controlled-route coherence, pure and mixed duality, the short-time coefficient, the hard/soft variance identity, the Duhamel enclosure, and scalar-phase accounting. Its finite oscillator modes are explicitly assigned comparison data. Their tests neither replace the continuum smearing nor certify its spectral measure; the continuum results above follow from the stated form-domain and hard-branch arguments.

8. A quadratic overlap enclosure for the declared continuum preparation

Provenance and scope. The quadratic refinement and robustness corollaries in this section are developed in campaign voli_chvi_foundation1_20261017 (H002700), with two native SLC-GEN5-R1 operator-word checks. They use the retained Chapter VI Hamiltonian, exact hard-eigenbranch preparation and continuum form construction. They do not replace that preparation by bare source levels, change its hard–soft split, or assert a new native certificate or a laboratory preparation. Route interpretation continues to use the chapter's declared proper-time controlled evolution.

8.1 Hamiltonian, preparation and operator-domain hypotheses

Retain the complete sixteen-state source and both continuum polarizations:

$$\begin{aligned} H_0 &= \varepsilon(S+Q_L+Q_R+Q_LQ_R)+H_f-\varepsilon(D\phi_+ + C\phi_\times) \\ &\quad +\varepsilon^2\mu(D^2+C^2), \\ D &= Q_L-Q_R, \quad C=iSD, \quad \mu=(4R)^{-1}. \end{aligned} \tag{70}$$

The result below concerns one invariant, separately prepared lane $a<b$, with signed $d=q_a-q_b<0$, $g=\varepsilon d$, and $e=q_a+q_b+q_aq_b$. The restriction of this same generator is

$$H_0 = \varepsilon eI+\varepsilon\sigma_x+H_f-g(\sigma_z\phi_+ +\sigma_y\phi_\times)+2g^2\mu I. \tag{71}$$

The lane restriction specifies the preparation sector; it does not discard the remaining source sectors from the model. No claim about an arbitrary superposition of different lanes is made here.

Use the chapter's declared split $\lambda = 64\Delta$ and factor the field into hard and soft Fock factors, including both polarizations in each. Let H_{ref} retain the hard interaction, free soft field, even source energy and the *same full counterterm* $2g^2\mu I$ as H_0. Thus their closed quadratic forms have difference

$$\begin{aligned} V_s &= -g(\sigma_z\phi_{+,s}+\sigma_y\phi_{\times,s}), \\ v_s &= \int_0^\lambda |u_R(\omega)|^2\,\mathrm{d}\omega \le v_s^{\mathrm{up}} = \frac{R^4\lambda^6}{675\kappa\pi}. \end{aligned} \tag{72}$$

In particular, no counterterm is subtracted a second time. The signed g remains in the interaction; its square below comes from a norm.

Both Hamiltonians are the self-adjoint, semibounded operators associated with the retained closed forms on the common form domain $\mathscr{Q}=D(H_f^{1/2})$, with the finite source factor understood. The full ultraviolet smearing need not be square integrable. The soft smearing in Eq. (72) is square integrable, and the chapter's weighted field estimates define the common forms. We do not treat V_s as a bounded operator on the full Fock space or subtract unbounded operators on an unspecified domain.

Let the normalized orthogonal exact hard eigenbranches, including the matched scalar terms, give

$$\begin{aligned} \psi_k &= \Phi_k\otimes|0_s\rangle \in D(H_{\mathrm{ref}}), \\ H_{\mathrm{ref}}\psi_k &= E_k\psi_k, \\ \Psi_0 &= \frac{\psi_0+\psi_1}{\sqrt{2}}, \\ \Omega &= E_1-E_0>0. \end{aligned} \tag{73}$$

Here $|0_s\rangle$ is the joint soft vacuum of both polarizations. Set

$$\begin{aligned} U(t) &= e^{-itH_0}, \\ R(t) &= e^{-itH_{\mathrm{ref}}}, \\ b^2 &= 2g^2v_s, \\ \bar{E} &= \frac{E_0+E_1}{2}. \end{aligned} \tag{74}$$

The symbol $R(t)$ denotes the reference propagator, whereas the scalar R in the form factor is the retained source radius.

For every real u, the form action of V_s on $R(u)\Psi_0$ is represented by the Hilbert-space vector $f(u)=V_sR(u)\Psi_0$. It lies in the one-soft-photon sector and is continuous in u. The two polarization insertions are orthogonal, while σ_z and σ_y preserve the source norm. Consequently

$$\begin{aligned} \|f(u)\|^2 &= b^2, \\ \langle\Psi_0, R(v)f(u)\rangle &= 0 \quad (u,v\in\mathbb{R}). \end{aligned} \tag{75}$$

The last identity uses the zero-soft-photon reference orbit, not a truncation of the actual evolution.

For completeness, the representation theorem for the common closed form gives, against every test vector in $\mathscr{Q}$,

$$\begin{aligned} R(u)\Psi_0 &\in D(H_0), \\ H_0R(u)\Psi_0 &= H_{\mathrm{ref}}R(u)\Psi_0+f(u). \end{aligned} \tag{76}$$

The right side is Hilbert-space continuous, since the reference orbit is a finite sum of eigenvectors and f is continuous. These are sufficient strong-Duhamel hypotheses. At $u = 0$ they also give

$$\begin{aligned} \langle H_0\rangle_{\Psi_0} &= \bar{E}, \\ \sigma_H^2 := \|(H_0-\bar{E})\Psi_0\|^2 &= \frac{\Omega^2}{4} + b^2. \end{aligned} \tag{77}$$

This uses $\Psi_0 \in D(H_0)$; neither $D(H_0^2)$ nor a fourth energy moment is assumed. No numerical value of $\bar{E}$ is supplied here.

8.2 The scalar cancellation

Theorem (quadratic scalar Duhamel enclosure). Under Eqs. (70)– (76), define

$$\begin{aligned} \Gamma_0(s) &= \langle \Psi_0, U(s)\Psi_0\rangle, \\ \Gamma_{\mathrm{ref}}(s) &= e^{-i\bar{E}s}\cos(\Omega s/2). \end{aligned} \tag{78}$$

Then, for every real s,

$$\boxed{|\Gamma_0(s) - \Gamma_{\mathrm{ref}}(s)| \le \frac{b^2 s^2}{2} = g^2 v_s s^2 \le q(s) := g^2 v_s^{\mathrm{up}} s^2.} \tag{79}$$

The actual propagator on the left retains the complete continuum and all photon sectors. The statement concerns the scalar survival amplitude; it does not replace the first-order bound on the state-vector difference.

Proof. Write $\mathscr{E}(t) = U(t) - R(t)$. Equation (76) permits differentiating $U(t-u)R(u)\Psi_0$ and integrating in the Hilbert space. With the oriented integral for negative t,

$$\begin{aligned} \mathscr{E}(t)\Psi_0 &= -i\int_0^t U(t-u) f(u)\,\mathrm{d}u, \\ \|\mathscr{E}(t)\Psi_0\| &\le b|t|. \end{aligned} \tag{80}$$

For $t \ge 0$, insert the second identity in Eq. (75) into the scalar matrix element to obtain

$$\langle \Psi_0, \mathscr{E}(t)\Psi_0\rangle = -i\int_0^t \langle \Psi_0, \mathscr{E}(t-u) f(u)\rangle\,\mathrm{d}u. \tag{81}$$

Here $\mathscr{E}(v)^* = \mathscr{E}(-v)$, so Cauchy–Schwarz and Eq. (80) give

$$\begin{aligned} |\langle \Psi_0, \mathscr{E}(t)\Psi_0\rangle| &\le \int_0^t \|\mathscr{E}(-(t-u))\Psi_0\|\,\|f(u)\|\,\mathrm{d}u \\ &\le b^2 \int_0^t (t-u)\,\mathrm{d}u = \frac{b^2 t^2}{2}. \end{aligned} \tag{82}$$

For negative t, self-adjointness gives the complex conjugate scalar difference at $-t$, with the same modulus. Finally, Eq. (73) evaluates the reference matrix element as the displayed cosine. This establishes Eq. (79). □

The chapter's previous radius was $\varepsilon_s = |s|\sqrt{2g^2 v_s^{\mathrm{up}}}$. The new radius is $q(s) = \varepsilon_s^2/2$. Both bounds remain valid; one may use $\min\{2, \varepsilon_s, q(s)\}$ at long times. This improvement does not certify an exact zero, a revival, a phase-clock inverse, or an extended physical two-arm coupling.

8.3 Preparation error and timing uncertainty

Corollary (full-marker preparation error). Let ρ be a normalized density operator on the *same complete* source–field Hilbert space, initially uncorrelated with the route system, and suppose

$$T(\rho, |\Psi_0\rangle\langle\Psi_0|) := \tfrac{1}{2}\|\rho - |\Psi_0\rangle\langle\Psi_0|\|_1 \le \delta_{\mathrm{prep}}. \tag{83}$$

For the unchanged actual generator, set $\Gamma_\rho(s) = \mathrm{Tr}(\rho U(s))$. Then

$$|\Gamma_\rho(s) - e^{-i\bar{E}s}\cos(\Omega s/2)| \le q(s) + 2\delta_{\mathrm{prep}}. \tag{84}$$

Proof. Trace/operator-norm duality and $\|U(s)\| = 1$ bound the change of amplitude by $2\delta_{\mathrm{prep}}$; add Eq. (79). □

The factor two follows from the stated trace-distance convention. Closeness of only the source marginal does not supply this full-marker preparation hypothesis. Trace-distance closeness also does not, by itself, give a finite energy variance for ρ. Neither inference is needed for Eq. (84).

Corollary (centered-energy timing bound). Let s be the nominal proper-time difference in the chapter's dimensionless chart and let the actual difference be $s+\xi$, with $|\xi| \le a$. Define the centered amplitude

$$G_\rho(t) = e^{i\bar{E}t}\Gamma_\rho(t) = \mathrm{Tr}[\rho e^{-i(H_0-\bar{E})t}]. \tag{85}$$

Then

$$\boxed{|G_\rho(s+\xi) - \cos(\Omega s/2)| \le q(s) + 2\delta_{\mathrm{prep}} + \sigma_H a.} \tag{86}$$

A certified upper bound for σ_H may replace σ_H. If the two route ages have errors $|\xi_j| \le a_j$, take $\xi = \xi_1 - \xi_2$ and $a = a_1 + a_2$; a common equal age error cancels.

Proof. For $A = H_0 - \bar{E}$, the spectral theorem and $|e^{-ix} - 1| \le |x|$ give

$$\|(e^{-iA\xi} - I)\Psi_0\| \le |\xi|\,\|A\Psi_0\| = \sigma_H|\xi|. \tag{87}$$

Consequently $|G_0(s+\xi) - G_0(s)| \le \sigma_H|\xi|$. Compare $G_\rho(s+\xi)$ first with $G_0(s+\xi)$, then with $G_0(s)$, and finally with the reference cosine. The three errors are bounded by $2\delta_{\mathrm{prep}}$, $\sigma_H a$ and $q(s)$, respectively. Thus only the exact preparation's variance is used. □

Centering is an explicit energy-phase assignment, not an instruction to discard an observable route phase. In raw amplitudes the conclusion is

$$\left|\Gamma_\rho(s+\xi) - e^{-i\bar{E}(s+\xi)}\cos(\Omega s/2)\right| \le q(s) + 2\delta_{\mathrm{prep}} + \sigma_H a. \tag{88}$$

If one instead compares with the *nominal* raw reference $e^{-i\bar{E}s}\cos(\Omega s/2)$, add the energy-phase error

$$2|\cos(\Omega s/2)|\,|\sin(\bar{E}\xi/2)| \le |\cos(\Omega s/2)|\min\{2, |\bar{E}|a\}. \tag{89}$$

Thus variance controls the centered timing error and the modulus error; a numerical absolute-phase bound additionally needs the mean energy or a specified compensating control phase. No numerical $\bar{E}$ is inferred from the gap Ω.

Under a common scalar reassignment $H_0 \mapsto H_0 + \beta I$ and $H_{\mathrm{ref}} \mapsto H_{\mathrm{ref}} + \beta I$, G_ρ, σ_H and q are unchanged, whereas $\Gamma_\rho(t) \mapsto e^{-i\beta t}\Gamma_\rho(t)$. Preserving a complete controlled-route amplitude requires the corresponding external action assignment $S_{\mathrm{ext},j} \mapsto S_{\mathrm{ext},j} + \hbar\beta\eta_j$ at the actual route ages. Even energy and counterterms remain counted once.

8.4 Modulus and phase disks

Corollary (readable amplitude and phase enclosure). Suppose any of the preceding estimates gives a complex disk $|z-c| \le r$ for a normalized amplitude z. Writing $m = |c|$,

$$\max\{0, m-r\} \le |z| \le \min\{1, m+r\}. \tag{90}$$

If $r < m$, the amplitude is nonzero and its circular phase distance obeys

$$\mathrm{dist}_{\mathbb{S}^1}(\arg z, \arg c) \le \arcsin(r/m). \tag{91}$$

Proof. The modulus bounds are the triangle and reverse-triangle inequalities, together with $|z| \le 1$. Rotate c to $m > 0$. For $r < m$ the disk lies in the open right half-plane; the distance from m to the ray through z is $m|\sin\theta| \le r$ with $|\theta| < \pi/2$. This gives Eq. (91). □

For exact timing use $c = e^{-i\bar{E}s}\cos(\Omega s/2)$ and $r = q(s) + 2\delta_{\mathrm{prep}}$. For the centered jitter estimate use $c = \cos(\Omega s/2)$ and $r = q(s) + 2\delta_{\mathrm{prep}} + \sigma_H a$. The modulus enclosure is then also an enclosure of the raw amplitude's modulus, and an ideal recombiner has visibility $2\sqrt{p_1 p_2}|z|$ under its stated assumptions. For the raw jittered phase, restore $-\bar{E}(s+\xi)$ explicitly. If the cosine is negative its reference phase includes π; at its zeros, or whenever $r \ge m$, this disk yields no phase enclosure. A valid local phase branch is therefore confined to an interval on which the disk excludes zero. Additional beam-splitter, routing and recombination phases must still be included in an experimental fringe.

8.5 Six-lane quadratic scalar-overlap bounds

The retained foundation native return evaluates the scalar cancellation above at the original cycle endpoint, on the same continuum preparation:

Lane	Previous linear remainder upper	New quadratic remainder upper
01	$1.758036931864852 \times 10^{-12}$	1.545347×10^{-24}
02	$3.643671179298703 \times 10^{-12}$	6.638170×10^{-24}
03	$4.535709727166836 \times 10^{-11}$	1.028634×10^{-21}
12	$1.885634247433851 \times 10^{-12}$	1.777809×10^{-24}
13	$4.359906033980351 \times 10^{-11}$	9.504391×10^{-22}
23	$4.171342609236966 \times 10^{-11}$	8.700050×10^{-22}

This amplitude error is distinct from the linear state-vector distance and the later preparation, transport and apparatus errors.

9. Preparation, timing, and causal retention

The purpose of this section is to specify the common operands for the next source-only decoder and detector-to-record calculations. Numerical laboratory values are not assigned to an unspecified preparation, control, or memory. The estimates below apply to explicitly declared channels and states and can be filled with source-derived bounds as those constructions are completed.

9.1 A preparation specification with retained failure

Let $\rho_* = |\Psi_0\rangle\langle\Psi_0|$ denote the exact Chapter VI hard-eigenbranch/soft-vacuum preparation in one declared lane. A physical preparation must return a complete instrument with a ready flag and all failure flags. For each operational input label j, write

$$\widehat{\rho}_j = (1-f_j)|\mathrm{ready}\rangle\langle\mathrm{ready}| \otimes \rho_j^{\mathrm{r}} + f_j|\mathrm{fail}\rangle\langle\mathrm{fail}| \otimes \rho_j^{\mathrm{f}}. \tag{92}$$

Multiple failure outcomes can be retained as separate blocks. Specify the ready-branch trace-distance error

$$\delta_{\mathrm{prep},j} = \tfrac{1}{2}\|\rho_j^{\mathrm{r}} - \rho_{*,j}\|_1. \tag{93}$$

When $\rho_{*,j}$ is pure, a fidelity floor $\langle\Psi_{*,j}|\rho_j^{\mathrm{r}}|\Psi_{*,j}\rangle \ge 1-\zeta_j$ gives $\delta_{\mathrm{prep},j} \le \sqrt{\zeta_j}$ by the standard trace-distance/fidelity relation [24]. This requires fidelity with the full specified source–field preparation; a source marginal fidelity alone does not certify the joint field state.

If failure and abstention are counted as unsuccessful decoding, the complete attempt law obeys the exact identity

$$e_j^{\mathrm{all}} = f_j + (1-f_j)e_j^{\mathrm{r}}. \tag{94}$$

Consequently $f_j \le \bar{f}$ and $e_j^{\mathrm{r}} \le \bar{e}$ imply $e_j^{\mathrm{all}} \le \bar{f} + (1-\bar{f})\bar{e}$. Conditioning on the ready flag defines a different, explicitly heralded protocol. Its success probability and retained failed attempts remain part of the resource account. A single experiment must choose whether it reports the all-attempt or ready-conditional record criterion.

9.2 A channel error budget that permits correlations

For each label j, let Φ_k and $\widetilde{\Phi}_k$ be reference and actual trace-preserving channels for the successive writing, switching, storage, transport, access, and decoding stages. Classical flags are included in their outputs. Registers needed to retain environmental correlations are included in the stage input. The allowed states must be rich enough that each bound below holds for the states on which that stage is actually used; if an arbitrary reference is retained, use the corresponding bound with that reference. Define

$$\delta_k = \sup_{j,t,\omega\in\mathscr{S}_{k,j,t}} \tfrac{1}{2}\|[(\widetilde{\Phi}_k - \Phi_k)\otimes I](\omega)\|_1. \tag{95}$$

The time supremum is over the declared receiving interval. A half-diamond-norm bound is sufficient; a bound uniform on the relevant reachable states also suffices. An environment with memory is retained as part of the joint state; independent noise between stages or cells is not assumed.

Proposition 1 (Composed readout error)**.** *If the reference decoder has worst-label error at most $e_*(t)$ and all stage bounds apply on the declared time interval, then the actual ready-branch decoder satisfies*

$$\sup_{j,t} e_j^{\mathrm{r}}(t) \le \min\{1,\ \sup_t e_*(t) + \delta_{\mathrm{prep}} + \sum_k \delta_k\}. \tag{96}$$

Proof. Insert the reference preparation and each reference channel one at a time into the composition. The triangle inequality bounds the final trace distance by $\delta_{\mathrm{prep}} + \sum_k \delta_k$; subsequent trace-preserving channels contract each difference. The success effect has spectrum in $[0,1]$, so its probability changes by at most that trace distance. Apply the bound to every label and time, then take the supremum. This argument also applies to a channel whose output contains both a classical instrument label and the corresponding quantum state. □

Equation (94) then adds retained preparation failure when the all-attempt criterion is used. Do not count that same failure a second time inside an all-attempt channel-distance term.

9.3 Timing errors and the energy account

All internal ages use the Chapter VII dimensionless chart. Let the actual readout ages differ from the nominal ages by ξ_1, ξ_2. If $|\xi_j| \le a_j$, the overlap time difference changes by at most $a_1 + a_2$. The centered-amplitude estimate in the preceding overlap section uses the full prepared energy variance. It is insensitive to a common scalar energy reassignment only after the compensating external action phase is included. Absolute fringe phase requires the matched hard mean energy and the scalar control-action convention. A gap interval alone does not fix that absolute phase.

For a switched or noncommuting control Hamiltonian, replacing the entire experiment by an age shift is a separate assertion. Instead, a common-domain strong Duhamel estimate can bound a declared perturbation $\Delta K_j(t)$:

$$r_j \le \int_0^{t_f} \|\Delta K_j(u)\Psi_{\mathrm{ref},j}(u)\|\, du. \tag{97}$$

The forcing must exist and be integrable in Hilbert norm. The reference orbit and domain assumptions must be stated for the actual singular-source control law. A nominal pulse duration by itself supplies no value of r_j. For normalized pure outputs, the resulting state trace distance is at most $\min\{1, r_j\}$, so it contributes directly to the decoder error budget. For branch overlap it contributes $r_1 + r_2$ by the triangle inequality.

If the good-timing event has probability at least $1-\beta$ for every label, and the conditional good-event decoding error is at most e_{good}, then the unconditional error is bounded by

$$e \le \beta + (1-\beta)e_{\mathrm{good}}. \tag{98}$$

The conditional input on that event must satisfy the preparation and control bounds used for e_{good}. Independence between timing, preparation, and detector noise is not needed when the bounds are uniform or formulated on their joint conditioned state.

9.4 The specification to carry into the next calculation

The following ledger distinguishes a defined mathematical operand from an unassigned implementation value.

Operand	Meaning	Present input or next receiving construction
$\rho_*, \Omega, M, v_s^{\mathrm{up}}$	Joint reference and certified bounds	Retained Chapter VI/Chapter VII package; exact hard branches are defined.
$\bar{E}, \Phi_0, \Phi_1$	Absolute energy and explicit hard state operands	Recover certified hard-state data for numerical absolute phase and preparation.
$f_j, \delta_{\mathrm{prep},j}$	Failure probability and ready-state accuracy	Bound from a complete finite preparation instrument.
a_j, β, r_j	Age uncertainty, timing tail, control mismatch	Derive from the specified clock, switch, and actuator.
$\mathscr{N}_t$	Storage including correlated noise and read disturbance	Choose a physical memory and retained environment.
$\mathscr{A}_{\mathscr{D},t}$	Transport and permitted receiving access	State receiver region, causal arrival, and available controls.
$F_j(t), e_*(t)$	Fixed experiment's decoder and reference error	Next source-only quadrature benchmark, then actual detector bins.
$\mathscr{T}_{\mathrm{mem}}, \mathscr{T}_{\mathrm{read}}$	Memory and receiving windows	Use a causal onset and uniform retention/readout bounds.

For repeated reads, the channel sequence includes every earlier read and its retained output. Uniform single-read performance alone specifies no bound on a sequence of disturbed reads. Initial source–apparatus correlations, finite control references, and failure recipients enter the declared state and channels rather than being discarded at the composition step.

The interface to the later two-arm transport task is fixed now: one full source–field state, declared branch controls and receiving regions, both polarizations, all source sectors, even energy and counterterms, and the same preparation/clock chart. The next detector calculation can then return a complete conditional law and its error budget under the stated transport law; actual two-arm transport will supply its additional controlled discrepancy.

10. A source-only decoder for the retained interacting clock

Provenance and receiving assumptions. The decoder and bounds below are developed in campaign voli_chvi_source_decoder1_20261017 (H002701) from the retained Chapter VII clock observable and its Chapter VI route interpretation, with four native SLC-GEN5-R1 calls. The source equations are in `CHAPTER_I_MODEL.md`, Sections 2 and 5: the exact source-lane restrictions, $Z = (D+iC)/d$, the reference expectation $h(t) = Me^{-i\Omega t}$, and the actual bound $|z(t) - h(t)| \le E(t)$. The original signed coupling, both field polarizations, even source energy and counterterms remain in the actual evolution that supplies z. The ideal POVM at apparatus entry is source-only. During a finite physical readout

the same full Hamiltonian can continue to correlate source and field; an implemented joint instrument must then approximate this ideal POVM on the specified entry states. No direct field readout is prescribed. Section 12 supplies that finite control and pointer construction with its explicit drift allowance.

We retain the chapter's conditional route law: the complete marker starts in its declared exact hard-eigenbranch/soft-vacuum preparation Ψ_0, uncorrelated with the route label, and route j supplies $e^{-iH_0\eta_j}\Psi_0$. The source output is the partial trace over the *complete* field. No field fragment is measured by this decoder.

10.1 The quadrature embedded in all sixteen source states

Let $|q_a\rangle$ be the normalized ordered eigenvectors of the original four-state source matrix Q. For one chosen lane $a<b$, abbreviate $|ab\rangle = |q_a\rangle\otimes|q_b\rangle$ and define operators on $\mathbb{C}^4\otimes\mathbb{C}^4$ by

$$\begin{aligned} P &= P_{ab}^{\mathrm{lane}} = |ab\rangle\langle ab| + |ba\rangle\langle ba|,\\ Z_{ab} &= |ab\rangle\langle ab| - |ba\rangle\langle ba|,\\ X_{ab} &= PSP, \qquad Y_{ab} = iSZ_{ab}, \end{aligned} \tag{99}$$

where S is the original full source swap. On the ordered lane $(|ab\rangle,|ba\rangle)$ these are $I,\sigma_z,\sigma_x,\sigma_y$; Z_{ab}, X_{ab} and Y_{ab} vanish on the other fourteen source states. In particular Y_{ab} is Hermitian, $Z_{ab}^2 = Y_{ab}^2 = P$, and $Z_{ab}Y_{ab} + Y_{ab}Z_{ab} = 0$. The complex clock observable embedded in this lane is

$$\begin{aligned} Z_{ab}^{\mathrm{clk}} = Z_{ab} + iY_{ab} = \frac{P(D+iC)P}{d_{ab}},\\ d_{ab} = q_a - q_b < 0. \end{aligned} \tag{100}$$

Every displayed source operator acts as its tensor product with the identity on the field. Thus this is the original source observable, not an observable requiring joint access to a dressed hard-field state.

The projector P commutes with Q_L, Q_R, S, D, C and preserves the common energy form domain. It consequently reduces the retained form-sum Hamiltonian. The exact lane preparation stays in this lane under H_0. Writing its reduced source state at age t as $\rho_S(t)$, we therefore have

$$\begin{aligned} z(t) &= \mathrm{Tr}[\rho_S(t)Z_{ab}^{\mathrm{clk}}],\\ O_\chi &= \frac{e^{-i\chi}Z_{ab}^{\mathrm{clk}} + e^{i\chi}(Z_{ab}^{\mathrm{clk}})^*}{2} = \cos\chi\, Z_{ab} + \sin\chi\, Y_{ab},\\ O_\chi^2 &= P, \qquad \|O_\chi\| = 1, \qquad m_j(\chi) = \mathrm{Re}[e^{-i\chi}z(\eta_j)]. \end{aligned} \tag{101}$$

For an explicitly complete binary measurement, assign every state outside the nominated lane to label zero:

$$\begin{aligned} E_1(\chi) &= \frac{P - O_\chi}{2},\\ E_0(\chi) &= I_{16} - E_1(\chi). \end{aligned} \tag{102}$$

These are orthogonal projections and sum to I_{16}. On the lane they are $(P\mp O_\chi)/2$; on its complement $E_0 = I$ and $E_1 = 0$. The full binary observable is accordingly $E_0 - E_1 = (I_{16} - P) + O_\chi$, rather than O_χ on the complement. No population has been dropped, and no success-conditioned renormalization is used. For lane-supported states the conditional correct-label probabilities are

$$\begin{aligned} p(0|0) &= \frac{1+m_0(\chi)}{2},\\ p(1|1) &= \frac{1-m_1(\chi)}{2}. \end{aligned} \tag{103}$$

These probabilities and their worst-label bounds do not require balanced label priors.

10.2 General ages and a source-only discrimination witness

Proposition (quadrature contrast). For two exact lane-supported source outputs, let $z_j = z(\eta_j)$ and $D_S = \frac{1}{2}\|\rho_S(\eta_0) - \rho_S(\eta_1)\|_1$. Then

$$D_S \geq \frac{|z_0 - z_1|}{2}. \tag{104}$$

If $z_0 \neq z_1$, the quadrature maximizing $m_0 - m_1$ is $\chi = \arg(z_0 - z_1)$ modulo 2π. This maximizes contrast within the stated quadrature family; it is not an assertion that this family is the unrestricted Helstrom measurement or that it maximizes worst-label success for arbitrary unequal-radius outputs.

Proof. For any real χ, trace/operator-norm duality gives $|m_0 - m_1| \leq 2D_S\|O_\chi\| = 2D_S$. Maximizing $\mathrm{Re}[e^{-i\chi}(z_0 - z_1)]$ gives $|z_0 - z_1|$. On lane-supported inputs the completed binary measurement also attains the corresponding average success $1/2 + |z_0 - z_1|/4$ for balanced priors. □

For the reference expectations $h_j = Me^{-i\Omega\eta_j}$ with the retained $M > 0$, put $\bar\eta = (\eta_0 + \eta_1)/2$ and $v = \sin[\Omega(\eta_1 - \eta_0)/2]$. When $v \neq 0$, choose

$$\chi_* = \arg(h_0 - h_1) = \frac{\pi}{2} - \Omega\bar\eta + \begin{cases} 0, & v > 0,\\ \pi, & v < 0, \end{cases} \qquad (\mathrm{mod}\ 2\pi). \tag{105}$$

The reference quadrature means are then $+M|v|$ and $-M|v|$. For a predeclared calibrated implementation of this axis, the retained actual-clock bound implies

$$\begin{aligned} \min_j p(j|j) &\geq \frac{1 + M|v| - \max_j E(\eta_j)}{2},\\ D_S &\geq \max\left\{0, M|v| - \frac{E(\eta_0) + E(\eta_1)}{2}\right\}. \end{aligned} \tag{106}$$

The second bound follows directly from $|z_0 - z_1| \geq |h_0 - h_1| - E(\eta_0) - E(\eta_1)$. If the reference contrast vanishes, the displayed distance lower bound is zero and no orientation is selected by this contrast criterion. Choosing an axis from reference data before a trial does not presume knowledge of its unknown label. Parameter uncertainty in that chosen axis must be included in its calibration error.

10.3 A fixed-axis, quarter-cycle protocol

The following protocol avoids choosing its nominal axis from an absolute hard energy or from the measured trial output. Use

the retained chart

$$\begin{aligned} T_0 &= \frac{2\pi}{\Delta}, \\ \eta_0 &= \frac{T_0}{4}, \\ \eta_1 &= \frac{3T_0}{4}, \\ \chi &= -\frac{\pi}{2}, \\ d &= 1-\frac{\Omega}{\Delta}. \end{aligned} \tag{107}$$

The exact retained lane data satisfy $0<\Omega\le\Delta$, so $d\ge 0$. The new symbol d in Eq. (107) denotes the relative gap deficit; it is distinct from the signed source difference d_{ab} in Eq. (100). The ages describe the two candidate marker histories at a common apparatus entry, not two label-dependent readout triggers. The same rotation, pointer test and outcome assignment are used for both hypotheses. The measured lane quadrature is $O_{-\pi/2}=-Y_{ab}$. Its reference means, including their signs, are

$$\begin{aligned} m_{0,\mathrm{ref}} &= M\cos(\pi d/2), \\ m_{1,\mathrm{ref}} &= -M\cos(3\pi d/2). \end{aligned} \tag{108}$$

Thus the two *correct-label margins* are positive cosines, whereas the raw means have opposite signs. The retained comparison error gives

$$\begin{aligned} p(0|0) &\ge \frac{1+M\cos(\pi d/2)-E(\eta_0)}{2}, \\ p(1|1) &\ge \frac{1+M\cos(3\pi d/2)-E(\eta_1)}{2}. \end{aligned} \tag{109}$$

Let outward inputs satisfy

$$\begin{aligned} 0\le m_*\le M\le 1, \\ 0\le d\le d_*, \\ \pi\le\pi_*, \\ \Omega\le\Omega_*, \\ E(t)\le E_* \\ (0\le t\le T_0). \end{aligned} \tag{110}$$

Here m_* is a lower bound for the retained M, not a rounded display. For example $d_*\ge 1-\Omega_{\mathrm{lower}}/\Delta_{\mathrm{upper}}$ is safe with positive denominator and the stated $\Omega\le\Delta$ premise. E_* can be supplied by the retained cycle enclosure of $E(t)=\exp(4g^2v_s^{\mathrm{up}}t^2)-1$, which is nondecreasing on this interval. The signed g and the original split remain unchanged.

Set $\Theta_0=3\pi_* d_*/2$. Whenever $\Theta_0\le\sqrt{2}$,

$$\boxed{\varepsilon_{\mathrm{clock}} := \max_j[1-p(j|j)] \le \frac{1-m_*[1-\Theta_0^2/2]+E_*}{2}.} \tag{111}$$

Indeed $\cos x\ge 1-x^2/2$ for real x. Both angular offsets in Eq. (108) have modulus at most Θ_0. The condition makes the common lower cosine bound nonnegative, so replacing M by its lower bound preserves the inequality. Without that condition the same substitution need not be valid; the universal fallback is the trivial error bound one.

10.4 Timing, axis and preparation robustness

Allow actual ages $t_j=\eta_j+\xi_j$ with $|\xi_j|\le a$ and $0\le t_j\le T_0$, and let the actual quadrature axis be $\chi=-\pi/2+\zeta$ with $|\zeta|\le b$. A sufficient age-window condition is $a\le T_0/4$. These are dimensionless ages and angular axis errors; no SI time calibration is introduced. The signed reference means now obey

$$\begin{aligned} m_{0,\mathrm{ref}} &= M\cos(-\pi d/2+\Omega\xi_0+\zeta), \\ -m_{1,\mathrm{ref}} &= M\cos(-3\pi d/2+\Omega\xi_1+\zeta). \end{aligned} \tag{112}$$

Both correct-label angular errors are bounded by

$$\Theta = \frac{3\pi_* d_*}{2}+\Omega_* a+b. \tag{113}$$

Consequently, for $\Theta\le\sqrt{2}$,

$$\max_j[1-p(j|j)] \le \frac{1-m_*[1-\Theta^2/2]+E_*}{2}. \tag{114}$$

The timing and axis losses enter through a squared displacement from the reference antipodal extrema. The square includes the gap-deficit and cross terms; it is not a claim that each uncertainty contributes an independent additive quadratic error. This bound is on a source probability, so it needs no absolute hard mean energy $\bar{E}$ or absolute route amplitude phase. Indeed a common scalar shift of the generator multiplies each branch propagator by a scalar phase, which cancels in $U_j\rho U_j^*$ and hence in its source measurement probabilities. That invariance does not remove the scalar contribution from an interferometric comparison of the route amplitudes themselves.

Suppose an imperfect complete marker preparation ρ satisfies

$$\tfrac{1}{2}\|\rho-|\Psi_0\rangle\langle\Psi_0|\|_1\le\delta_{\mathrm{prep}}, \tag{115}$$

with the same generator and declared ages. Unitary evolution and partial trace do not increase this distance. Each completed-POVM outcome probability therefore changes by at most δ_{prep}: for an effect $0\le F\le I$, the trace-distance variational bound is $|\operatorname{Tr}[F(\rho-\sigma)]|\le T(\rho,\sigma)$. It follows that

$$\boxed{\begin{gathered}\varepsilon_{\mathrm{dec}}\le\min\left\{1,\frac{1-m_*[1-\Theta^2/2]+E_*}{2}+\delta_{\mathrm{prep}}\right\}, \\ \Theta\le\sqrt{2}.\end{gathered}} \tag{116}$$

This includes leakage into the other fourteen source states and source–field preparation errors covered by Eq. (115). The penalty is δ_{prep} for an outcome probability, rather than the $2\delta_{\mathrm{prep}}$ bound for a general unitary amplitude. If an apparatus construction subsequently bounds an additional per-label outcome-probability error by δ_{read}, it may be added to the right side before clipping at one. That implementation error is not established by the ideal measurement algebra alone. In particular, the finite-control Kraus operators may act jointly on source and field even when all external controls and the pointer couple only to the source. Their approximation to the ideal entry POVM, rather than an exact source–field factorization during the pulse, is the required estimate.

If Eq. (115) is established only on a heralded ready branch, its resulting error bound is conditional on that branch. With per-label failure probability at most f, counting a failure as unsuccessful gives the all-attempt bound $f+(1-f)\varepsilon$, where $\varepsilon \le 1$ is the complete ready-branch error including preparation and readout. The failure outcome remains in the instrument; this formula does not assert an additional implemented flag or silently discard unsuccessful attempts.

10.5 Rotation sign and the complete binary readout

For a general nominal axis let $R_\chi = e^{-i\chi S/2}$. The original swap satisfies $S^2 = I_{16}$ and commutes with P, while it anticommutes with Z_{ab} on the lane. Hence

$$\begin{aligned} R_\chi^* Z_{ab} R_\chi &= \cos\chi\, Z_{ab} + \sin\chi\, Y_{ab} = O_\chi, \\ E_1(\chi) &= R_\chi^* |ba\rangle\langle ba| R_\chi. \end{aligned} \tag{117}$$

In particular

$$\begin{aligned} R &= e^{+i\pi S/4}, \\ R^* Z_{ab} R &= -Y_{ab}, \\ E_1 &= R^* |ba\rangle\langle ba| R, \\ E_0 &= I_{16} - E_1. \end{aligned} \tag{118}$$

Thus rotating by R and testing for $|ba\rangle$ realizes the specified *ideal POVM*; every other outcome is assigned label zero. Equivalently, a three-valued Z_{ab} read after the rotation can be coarse-grained by assigning its zero eigenspace to label zero. An exchange-angle error $R = e^{i(\pi/4+\alpha)S}$ gives quadrature-axis error $\zeta = -2\alpha$; a bound on $|\alpha|$ must therefore be doubled when used as the axis budget b. Other errors caused by a concurrently running drift or field interaction need their own control estimate.

For a separately declared leakage-calibration experiment one may instead use the three effects

$$\begin{aligned} &R^* |ab\rangle\langle ab| R, \\ &R^* |ba\rangle\langle ba| R, \\ &I_{16} - P. \end{aligned} \tag{119}$$

The third effect is a failure/leakage outcome and the three sum to I_{16}. This is a different stated output alphabet; its existence as a POVM does not supply a second physical flag apparatus, and conditioning away its third outcome would require a separate probability account.

The present section supplies the complete source-only POVM and its worst-label error budget. Section 12 implements its rotation and eigenbasis test with finite bounded controls and a two-state pointer. The durable memory, causal receiving channel and microscopic control actuator form the next apparatus join.

11. Exact source-bound values for the six original lanes

The original source certificate supplies the exact bare gap $\Delta = 2/573049$ and $\varepsilon = 1/573049$. For each lane use its retained outward endpoints for Ω, M, c and K_{prep}. Define entirely rational enclosing inputs

$$\begin{aligned} d_* &= 1 - \Omega_{\mathrm{lower}} \frac{573049}{2}, \\ \pi_* &= \frac{\Delta_{\mathrm{upper}} T_{0,\mathrm{upper}}}{2}, \\ q_* &= g^2 v_s^{\mathrm{up}} T_0^2 \text{ (retained outward rational)}, \\ E_* &= 8 q_*. \end{aligned} \tag{120}$$

All six lanes have $0 \le 4q_* \le 1/2$, so $e^{4q_*} - 1 \le 8q_*$ by the positive exponential series and $e^x \le (1-x)^{-1}$ for $0 \le x < 1$. The exact q_* values are inherited from the foundation capsule; they use its original continuum soft moment and upper cycle endpoint. The certificate endpoints enclose the exact bare gap, making $\pi_* \ge \pi$. The exact gap is used in d_* rather than a rounded gap display. The independent check verifies all sign, interval and nonnegative-cosine conditions before making a bound.

The largest ideal-entry bound occurs in lane 03. For the budgeted example declare

$$\begin{aligned} \Omega_* a &= \frac{1}{100}, \\ b &= 0, \\ \delta_{\mathrm{prep}} &= \frac{1}{10000}, \\ r_{\mathrm{ctrl}} &\le \frac{1}{1000}. \end{aligned} \tag{121}$$

The phase allowance here is entirely an age/timing allowance; the nominal analyser axis is exact. The resulting worst-label ready-branch bound is

$$\begin{aligned} \varepsilon_{\mathrm{ready}} \le \min\Bigg\{ 1, \frac{1 - m_*[1 - (3\pi_* d_*/2 + 1/100)^2/2] + E_*}{2} \\ + \frac{1}{10000} + \frac{1}{1000} \Bigg\} \le 0.0011254539 \end{aligned} \tag{122}$$

across all six lanes. The age allowance lies inside the original entry window in every lane. These values specify a receiving model; they do not assign an achieved laboratory preparation, timing or pointer accuracy. An ideal blank pointer and its ideal final binary observation are assumed in this benchmark. Additional initialization/readout errors can instead be supplied as joint-state or outcome-probability bounds. Per-label ready failure at most f gives all-attempt error at most $f+(1-f)\varepsilon_{\mathrm{ready}}$, with the failure output retained.

12. Finite source controls and a two-state readout pointer

This construction specifies a source-selective receiving model. Its controls act on the original sixteen-state source and a finite pointer; the complete continuum field remains active. It supplies a finite mathematical implementation of the stated source effects with an explicit drift and switching-work budget. The actuator implementing these source operators, its timing and calibration, and any autonomous replacement for the driven controls remain physical apparatus assignments.

Lane	Ideal-entry error upper	Finite-pulse r upper	Budgeted error upper
01	$6.2305241 \times 10^{-10}$	0.00014722126	0.0011250007
02	2.6763754×10^{-9}	0.00015791109	0.0011250030
03	4.1473953×10^{-7}	0.00082431540	0.0011254539
12	$7.1677609 \times 10^{-10}$	0.00014772605	0.0011250008
13	3.8321098×10^{-7}	0.00079335659	0.0011254194
23	3.5077944×10^{-7}	0.00076019657	0.0011253839

Table 2: Outward bounds for each original lane. The ideal-entry column uses exact preparation and nominal ages. The finite-pulse column uses the explicit mathematical controls below. The budgeted column adds the declared timing, preparation and 10^{-3} control allowances in Eq. (121). These allowances are receiving inputs.

12.1 The retained source energy is enough for short pulses

In the declared prepared lane retain

$$\begin{aligned} H_0 &= \varepsilon e I + \varepsilon S + K, \\ K &= H_f + H_{\mathrm{int}} + cI \geq 0, \\ c &= 2g^2\mu. \end{aligned} \tag{123}$$

Here S is the original Swap restriction, $\|S\| = 1$, and the even energy, both physical polarizations, both counterterms, hard dressing, soft vacuum, and all field complements are retained. On the original common form domain the annihilation form estimate gives

$$|\langle H_{\mathrm{int}} \rangle| \leq 2\sqrt{c\langle H_f \rangle} \leq \langle H_f \rangle + c. \tag{124}$$

Consequently, with the lane scalar removed only from the comparison phase,

$$\begin{aligned} &L = H_0 - \varepsilon e I + \varepsilon I \geq 0, \\ &L \leq 2\varepsilon I + 2H_f + 2cI \quad \text{as quadratic forms.} \end{aligned} \tag{125}$$

The scalar $\varepsilon e - \varepsilon$ is still part of the physical generator and its elapsed phase; removing its common phase from this comparison does not subtract its energy or any relative route phase.

Precisely, the positivity and scalar subtraction in Eq. (125) apply on the selected invariant lane tensored with the full Fock space and pointer. They are not a global all-sixteen-state positivity claim after subtracting one lane's e. The actual H_0 and bounded source controls are defined on all sixteen source states. Swap and P_{ba} preserve each invariant source lane, so the finite-energy proof restricts to the reference input lane without discarding the other states from the physical dynamics.

Suppose the declared reference source–field input to the readout has $\langle H_f \rangle \leq F$ at its declared entry age. Adjoin the blank finite pointer. Let V be an ideal source–pointer unitary and B a self-adjoint contraction, both preserving the selected lane and commuting with H_f. The controls used below have these properties. Then for $\xi = V\psi$ the form bounds imply

$$\begin{aligned} \langle \xi, L\xi \rangle &\leq C, \\ \langle B\xi, LB\xi \rangle &\leq C, \\ C &= 2\varepsilon + 2F + 2c. \end{aligned} \tag{126}$$

The second expression is not normalized: $\|B\xi\| \leq 1$ and $\langle B\xi, H_f B\xi \rangle \leq F$. Commutation with H_f preserves its form domain and is used here; preservation of $D(H_0)$ by B or V is neither assumed nor needed. A mixed input is covered by a purification, with the same expectation bounds and an untouched reference.

The retained source estimate can be inserted before selecting pulse times. For example, if the declared input-age bound is

$$F \leq 2c + 2K_{\mathrm{prep}} + 2\varepsilon E(T_0), \tag{127}$$

then one may use the outward source-bound ceiling

$$C \leq C^{\mathrm{up}} = 2\varepsilon + 6c + 4K_{\mathrm{prep}} + 4\varepsilon E(T_0). \tag{128}$$

This substitution inherits the preparation, input-age window and definitions of its source bound. It does not assign a new measured accuracy or an energy second moment.

12.2 A bounded-control drift lemma

For $L \geq 0$ and any vector in its form domain, the spectral theorem gives

$$\|(e^{-itL} - I)\xi\|^2 = \int 2[1 - \cos(tE)]\, d\mathscr{P}_\xi(E) \leq 2|t|\langle \xi, L\xi \rangle. \tag{129}$$

Thus a finite first energy form suffices, even when $L\xi$ is not a Hilbert-space vector.

Let a pulse of duration $\tau > 0$ have actual generator

$$\begin{aligned} H(t) &= L + \lambda(t)B, \\ A_{\mathrm{abs}} &= \int_0^\tau |\lambda(t)|\, dt < \infty, \end{aligned} \tag{130}$$

with B as above. The finite pointer is degenerate during this modelled control, and $\lambda(t)$ is real. In the interaction picture put

$$\begin{aligned} B_H(t) &= e^{itL} B e^{-itL}, \\ V(t) &= \exp\left[-iB \int_0^t \lambda(u)\, du \right]. \end{aligned} \tag{131}$$

The ideal $V(t)$ commutes with H_f. For $\xi_t = V(t)\psi$, split

$$\begin{aligned} (B_H(t) - B)\xi_t = e^{itL} B (e^{-itL} - I)\xi_t \\ + (e^{itL} - I) B \xi_t. \end{aligned} \tag{132}$$

Equations (126) and (129) therefore give

$$\begin{aligned} \|(B_H(t) - B)V(t)\psi\| &\leq 2\sqrt{2tC}, \\ 0 \leq t &\leq \tau. \end{aligned} \tag{133}$$

All free factors in this calculation are unitary on the Hilbert space. No commutator $[L,B]$, derivative $LB\xi_t$, or operator-domain preservation is required.

The actual interaction-picture generator $\lambda(t)B_H(t)$ is bounded and strongly measurable with integrable norm. Its Dyson propagator exists on the full Hilbert space. Bounded-generator Duhamel, applied to the ideal reference vectors only, gives

$$\|(U_I(\tau)-V(\tau))\psi\| \le 2\int_0^\tau |\lambda(t)|\sqrt{2tC}\,dt \le 2A_{\rm abs}\sqrt{2\tau C}. \tag{134}$$

The actual Schrödinger propagator is $U(\tau)=e^{-i\tau L}U_I(\tau)$. Adding its final free factor, and using the form bound on $V(\tau)\psi$, yields the finite-pulse estimate

$$\boxed{\|(U(\tau)-V(\tau))\psi\| \le (1+2A_{\rm abs})\sqrt{2\tau C}.} \tag{135}$$

This is a state-specific finite-energy estimate. For rectangular pulses, $L+\lambda B$ is self-adjoint on $D(L)$ by bounded perturbation and has the same form domain after a finite scalar shift. Its unitary preserves that form domain. This also makes the endpoint mean energies and bounded switching payments well defined for the stated input. The proof of Eq. (135) itself uses bounded Dyson evolution and the reference form estimates, not an assumed invariance of $D(L)$.

12.3 Exact ideal rotation and phase-free pointer gate

Take the ideal source rotation

$$R = e^{i\pi S/4}. \tag{136}$$

It is generated by $B_{\rm rot}=S$ with signed area $\int\lambda_{\rm rot}=-\pi/4$, hence $A_{\rm rot}=\pi/4$. The full source operator is the original six-lane Swap with its four unforced directions; it is not replaced by a bare Hamiltonian during the pulse.

Let P_{ba} be the declared source-only projector onto the ordered state selected by the receiver, and let $X_{\rm ptr}$ be the pointer Pauli matrix. Assign the bounded projector generator

$$B_{\rm gate} = P_{ba}\otimes\frac{I-X_{\rm ptr}}{2}. \tag{137}$$

At area $\int\lambda_{\rm gate}=\pi$,

$$\begin{aligned} G = e^{-i\pi B_{\rm gate}} &= I-2B_{\rm gate} \\ &= (I-P_{ba})\otimes I + P_{ba}\otimes X_{\rm ptr}. \end{aligned} \tag{138}$$

Thus a blank pointer $|0\rangle$ is flipped precisely on the selected source subspace, with no extra conditional phase. Both generators have norm at most one and commute with H_f. Their finite source–pointer matrices act on all 16×2 coordinates while the entire field remains in the dynamics.

For reference, put $P^{\rm ord}_{ab}=|ab\rangle\langle ab|$. In the ordered prepared-lane basis $|ab\rangle,|ba\rangle$ define $Z=P^{\rm ord}_{ab}-P_{ba}$, $X=S$, and $Y=-i|ab\rangle\langle ba|+i|ba\rangle\langle ab|$. Then

$$\begin{aligned} P_{ba} &= \frac{I-Z}{2}, \\ R^\dagger P_{ba} R &= \frac{I+Y}{2}. \end{aligned} \tag{139}$$

The two corresponding ideal signed pointer means are $\langle Z\rangle = 1-2P(1)$ without R and $\langle Y\rangle = 2P(1)-1$ after R. These signs belong to this explicit ordered basis. Any source-current convention used elsewhere must be translated by its actual source operator map.

12.4 Two-pulse error, finite switching work and event time

Let $U_{\rm rot},U_{\rm gate}$ be the actual rectangular pulses and R,G the ideal ones. Unitary invariance and one hybrid intermediate state give

$$\begin{aligned} &\|U_{\rm gate}U_{\rm rot}\psi - GR\psi\| \\ &\quad\le \|(U_{\rm rot}-R)\psi\| + \|(U_{\rm gate}-G)R\psi\| \\ &\quad\le r, \\ r &= \left(1+\frac{\pi}{2}\right)\sqrt{2C\tau_{\rm rot}} + (1+2\pi)\sqrt{2C\tau_{\rm gate}}. \end{aligned} \tag{140}$$

The second pulse is tested on the ideal state $R\psi$, whose H_f expectation is still bounded by F. The estimate does not require a new field-energy estimate on the actual intermediate state. A pointer-only readout omitting the rotation uses the second term alone.

For any final pointer event, the probability discrepancy is at most r, because pure-state trace distance is no larger than the vector difference. Partial trace and purification give the same bound for a mixed input. A binary pointer value in $\{-1,+1\}$ consequently has mean error at most $2r$. Pointer initialization and detector errors, when present, must be added using their declared all-outcome probability or joint-state bounds. These inequalities introduce no additional measurement rule.

If the actual preparation is known only to satisfy $T(\rho_{\rm act},\rho_{\rm ref})\le\delta_{\rm prep}$, apply the same full actual pulse and pointer channel to both states. Trace-distance contractivity adds at most $\delta_{\rm prep}$ to each output probability error. The reference pulse estimate therefore gives

$$|P_{\rm act}(E)-P_{\rm ideal,ref}(E)| \le \delta_{\rm prep}+r. \tag{141}$$

No field-energy or form-domain assumption is thereby made on $\rho_{\rm act}$, which may have a component outside the reference lane. The C bound and the pulse-drift proof concern the ideal reference states. Finite source-energy claims for an actual preparation, when needed for a resource account, require its own supplied energy bound.

Calibrated areas use their actual absolute areas. The nominal analyser angle is $\chi=-\pi/2$, corresponding to the source area $\chi/2=-\pi/4$. If a rectangular rotation has $|\delta\chi|\le b$, its frozen-unitary error relative to R is at most $b/2$, and its absolute area is at most $\pi/4+b/2$. If the pointer area is $\pi+\delta_g$ with $|\delta_g|\le u$, its frozen-unitary error relative to G is at most u and its absolute area is at most $\pi+u$. A sufficient full bound is therefore

$$\begin{aligned} r_{\rm cal} \le{}& \frac{b}{2}+u \\ &+\left(1+\frac{\pi}{2}+b\right)\sqrt{2C\tau_{\rm rot}} \\ &+(1+2\pi+2u)\sqrt{2C\tau_{\rm gate}}. \end{aligned} \tag{142}$$

This follows from $\|e^{-i\delta B}-I\|\le|\delta|$ for a self-adjoint contraction and the same hybrid comparison. A net-area uncertainty

does not bound the absolute area of a pulse with uncontrolled sign reversals; that case must supply A_{abs} itself. Nominal-area feasibility values use $b = u = 0$ in the pulse estimate. A separate timing or accumulated-phase allowance, for example $|\Omega\,\delta\eta|$, is not an analyser-axis error unless its declared control map makes it one. General calibrated work bounds likewise replace the nominal areas in Eq. (145) by these actual absolute-area ceilings.

There are two equivalent ways to allocate the rotation-axis error, and they must not be added together. Equation (142) compares every pulse with the *nominal* ideal rotation; the separate decoder phase/age error then excludes the same b. Alternatively compare the finite rotation with its *actual frozen* area and include b in the decoder's ideal-axis phase allowance. The finite-pulse part in that allocation is

$$\begin{aligned} r_{\text{actual axis}} \le u + \left(1 + \frac{\pi}{2} + b\right)\sqrt{2C\tau_{\mathrm{rot}}} \\ + (1 + 2\pi + 2u)\sqrt{2C\tau_{\mathrm{gate}}}, \end{aligned} \tag{143}$$

with no additional $b/2$. Both allocations use the actual absolute areas in their drift bounds. A nominal-axis benchmark with $b = 0$, whose stated phase allowance comes entirely from age/timing, has no such axis payment.

For a desired positive vector-error allowance $r_{\mathrm{rot}} + r_{\mathrm{gate}} = r_*$, one sufficient finite choice is

$$\begin{aligned} \tau_{\mathrm{rot}} &\le \frac{r_{\mathrm{rot}}^2}{2C^{\mathrm{up}}(1+\pi/2)^2}, \\ \tau_{\mathrm{gate}} &\le \frac{r_{\mathrm{gate}}^2}{2C^{\mathrm{up}}(1+2\pi)^2}. \end{aligned} \tag{144}$$

Choose strictly positive times. Their finite rectangular control heights are $\pi/(4\tau_{\mathrm{rot}})$ and π/τ_{gate}; tightening the error allowance charges these larger amplitudes explicitly.

At each on/off jump the changed part of the Hamiltonian is a bounded operator. The sum of absolute mean switching payments is therefore at most

$$\begin{aligned} W_{\mathrm{abs}}^{\mathrm{switch}} &\le 2\left(\frac{A_{\mathrm{rot}}}{\tau_{\mathrm{rot}}} + \frac{A_{\mathrm{gate}}}{\tau_{\mathrm{gate}}}\right) \\ &= \frac{\pi}{2\tau_{\mathrm{rot}}} + \frac{2\pi}{\tau_{\mathrm{gate}}}. \end{aligned} \tag{145}$$

Directly changing from one rectangular pulse to the next satisfies this same upper bound by the triangle inequality. This is an absolute bound on the declared driven switching work, not a deduction of autonomous controller capital. Initial source/field energy, the supplied blank pointer, actuator resources, retained outcomes, and any eventual reset remain separate recipients. Time and energy use the original model chart; no new SI assignment is supplied.

The selected projector control need not commute with the original source–field charge. Its actuator and phase/action reference are part of this external receiving model. The switching-work estimate does not by itself supply an autonomous charge-conserving actuator or its microscopic SAM realization.

The state to which the ideal readout refers is the state at pulse entry. Its ideal H_f bound is used throughout the comparison because the reference controls preserve H_f. The actual classical pointer result is available only after the latency $\tau_{\mathrm{rot}} + \tau_{\mathrm{gate}}$ and its subsequent registered readout. Calling it the source value at completion would require the corresponding event-time and phase-advance map. If the complete physical readout must lie inside the original certified window, choose its entry and positive pulse durations accordingly. No rate certificate is extended merely by omitting this finite latency.

The same fixed rotation, pointer gate, durations and output convention are used for all declared input-age hypotheses $\rho(\eta_j)$. The unknown label is not supplied to a control schedule. Its physical preparation and the two-arm arrival map are separate receiving-model data; a label-dependent ideal timing choice cannot be substituted for a single implemented decoder.

13. The instrument and an explicit finite pulse choice

Write $P_1 = P_{ba}$ and $P_0 = I_{16} - P_{ba}$. In the ideal frozen-control model with a blank pointer, the joint source–field Kraus operators and the measurement channel are

$$\begin{aligned} K_y &= (P_y R) \otimes I_f, \\ \mathscr{J}_y(\rho) &= K_y \rho K_y^*, \\ \sum_y K_y^* K_y &= I, \\ \mathscr{J}(\rho) &= \sum_{y=0}^{1} |y\rangle\langle y| \otimes \mathscr{J}_y(\rho). \end{aligned} \tag{146}$$

Thus $K_y^* K_y = E_y \otimes I_f$. The post-readout source can be disturbed by the projector; the field is retained in each output rather than replaced by a scalar marker. With the actual full drift and pulses, $\widetilde{K}_y = \langle y|U_{\mathrm{gate}}U_{\mathrm{rot}}|0\rangle$ is in general a joint source–field operator. Unitarity still gives $\sum_y \widetilde{K}_y^* \widetilde{K}_y = I$. The complete ideal and actual postmeasurement channels, with outcomes and quantum output retained, are within trace distance r on the reference input because the corresponding premeasurement pure states are within vector distance r. Applying the same channel to a preparation within trace distance δ_{prep} adds at most δ_{prep} to this joint-output distance. The two instruments therefore carry an all-outcome bound, not only a contrast estimate.

Use rectangular nominal pulses of equal finite duration

$$\begin{aligned} \tau_{\mathrm{rot}} = \tau_{\mathrm{gate}} = \tau = \frac{1}{32768}, \\ C^{\mathrm{up}} = 2\varepsilon^{\mathrm{up}} + 6c^{\mathrm{up}} + 4K_{\mathrm{prep}}^{\mathrm{up}} + 4\varepsilon^{\mathrm{up}}E_*. \end{aligned} \tag{147}$$

The exact native comparison evaluates

$$r^2 \le 2C^{\mathrm{up}}\tau\left(2 + \frac{5\pi_*}{2}\right)^2 \le 6.7949588 \times 10^{-7} < 10^{-6} \tag{148}$$

for all six lanes. In particular $r \le 0.00082431540$ across the six lanes, leaving the declared 10^{-3} control allowance sufficient for the nominal mathematical pulses. Their strengths and absolute

switching-work ceiling are finite and explicitly charged:

$$\begin{aligned} |\lambda_{\mathrm{rot}}| &\le \frac{\pi_*}{4\tau} < 25735.928, \\ |\lambda_{\mathrm{gate}}| &\le \frac{\pi_*}{\tau} < 102943.71, \\ W_{\mathrm{abs}}^{\mathrm{switch}} &\le \frac{5\pi_*}{2\tau} < 257359.28. \end{aligned} \tag{149}$$

These are original model units, without an assigned SI conversion or an observed control calibration. They display the strength/work cost of this short-pulse construction. An actuator/reference implementing the spectral projector and exchange drive must supply its charge, phase, timing and energy resources. The two pulses take the positive latency $1/16384$ before subsequent pointer registration. The ideal observable remains the observable at pulse entry.

14. A written bit on the actual nonuniform detector pointer

The completed Chapter IV detector supplies an actual continuum measurement and a finite uncentered second moment. Its positive intrinsic cap mean remains a separate source-to-field transport operand. The construction below uses the Chapter VI source decoder to write a calibrated displacement onto that actual detector pointer. The unknown continuum signal is retained in the background noise, rather than assigned a positive mean. Thus this is an explicit added bit-to-meter receiving law, not a certificate that the unmodified radiation alone distinguishes the two source ages.

The original sixteen source states, six signed lanes, both field polarizations and counterterms, hard dressing, soft input and all angular complements remain. One nominated lane and one of the actual local pointer coordinates suffice for the binary receiving protocol. If both compatible physical pointers are operated, the other pointer and its outcomes are retained; no independence between their records is inferred.

14.1 Re-key the field energy after the source decoder

Let the complete source-decoder output retain its two-state bit B and the source–field system. It can be correlated or entangled across these recipients. In the selected reference lane use

$$\begin{aligned} L &= H_0 - \varepsilon e I + \varepsilon I \ge 0, \\ H_f &\le 2L + 2cI, \\ c &= 2g^2\mu. \end{aligned} \tag{150}$$

These are the original completed-square form inequalities; the scalar subtraction is lane-specific. Suppose the reference decoder input has $\langle L\rangle \le C_{\mathrm{dec}}$ and its finite rectangular controls have total absolute switching-work bound W_{dec}. Their bounded Hamiltonian jumps and conservation between jumps imply

$$\begin{aligned} \langle L\rangle_{\mathrm{out}} &\le C_{\mathrm{dec}} + W_{\mathrm{dec}}, \\ E_{\mathrm{rec}} := \langle H_f\rangle_{\mathrm{out}} &\le 2(C_{\mathrm{dec}} + W_{\mathrm{dec}}) + 2c. \end{aligned} \tag{151}$$

The nonselective source-bit observation, or merely retaining the coherent bit, does not increase this source–field expectation. Do not condition this energy bound on a correct decoder outcome. All bit outcomes remain in the joint state. The old Chapter IV accepted-state energy and its numerical noise table are not substituted after the new driven decoder.

For an actual preparation supplied only within trace distance δ_{prep} of that reference, apply the same entire receiving channel to the two inputs and add δ_{prep} once at the final probability bound. No energy or form-domain assumption on the actual preparation follows from trace distance alone. Equation (151) and the ensuing second-moment calculation concern the finite-energy reference input and its complete, unselected decoder output.

14.2 The actual continuum gate on this new input

After the source bit is isolated, the actual nonuniform source–field propagator and detector gate act trivially on B. For a prescribed source-excluding local gate let X be its actual integrated field observable in the unmeasured joint Heisenberg picture. The original retained reflection and causal commutator statements give the scalar self-Magnus coefficient c_h. Write

$$\begin{aligned} C_f &\le 1.00301, \\ c_{\max} &= \frac{C_f^2 V_c}{2}, \\ |c_h| &\le c_{\max}. \end{aligned} \tag{152}$$

Here μ_c, V_c, D_c are the full compact-cap energy-dual, vacuum and half-energy squared norms, and $\mu_{\mathrm{sh}} = 1/(4R_s)$ is the original source-shell norm. The source radius R_s is distinct from the bit register B. Let t_* bound the actual new elapsed source-to-gate epoch, including the declared receiving chronology. The backward smooth-test estimate gives

$$\begin{aligned} \langle X^2\rangle &\le M_{\mathrm{rec}}, \\ M_{\mathrm{rec}} &= 6C_f^2\mu_c E_{\mathrm{rec}} + 3C_f^2 V_c \\ &\quad + 48g^2 C_f^2 n_+^2 \mu_{\mathrm{sh}} t_*^2 D_c, \qquad n_+ = \frac{2003}{2000}. \end{aligned} \tag{153}$$

The input may be mixed and correlated with B: the field estimate uses its unselected initial field energy and the unchanged bounded source-current norms. The bit is a spectator in this evolution. No final old-Fock energy, angular-vacuum factorization, Gaussian field state or positive cap mean is inserted. The signed g remains in the source law; its square appears only in this moment bound.

Let Q, P be the selected detector pointer with $[Q,P] = i$, and let kP^2, $k > 0$, be its declared kinetic Hamiltonian. The chosen k retains the Chapter IV field-form completion margin, or that margin is supplied by the additional quadratic stabilizer below. The gate has duration ℓ and normalized positive profile h. A small positive pointer mass coefficient alone does not stabilize the bilinear field gate.

For clarity, make the gate's form certificate explicit. In the static normal frame or the common moving frame used to construct the actual joint propagator, let $F_t \ge 0$ be the reference

field-energy form and suppose

$$\begin{gathered} H_{\mathrm{sf}}(t) \geq a_0 F_t - b_0 I, \\ a_0 > 0, \\ \|a_t(f)\psi\|^2 \leq \mu_* F_t[\psi]. \end{gathered} \tag{154}$$

The latter energy-dual bound is in the same frame, with $\Phi_t(f) = a_t(f) + a_t(f)^*$. The finite coefficients a_0, b_0, μ_* belong to that actual compact-profile common-form certificate. They are not a claim that its conjugating implementer preserves the old $Q(H_f)$. If a fixed old-energy frame with the stated lower bound is available, one may take $F_t = H_f$ and $\mu_* = \mu_c$; that substitution must be certified, rather than inferred from the initial-energy bound. On the product form domain, energy-dual Cauchy–Schwarz gives

$$|h\langle P\Phi_t(f)\rangle| \leq \frac{a_0}{2}\langle F_t\rangle + \frac{2\mu_*}{a_0}h^2\langle P^2\rangle. \tag{155}$$

Thus add a known smooth compact pointer control $s(t)P^2$, with

$$s(t) \geq \frac{2\mu_*}{a_0}h(t)^2. \tag{156}$$

The total gate form retains the lower bound $(a_0/2)F_t + kP^2 - b_0 I$; a slightly larger s gives a strict relative form margin for the common-form construction. This is a declared receiving actuator, not a change of the original source counterterms. The compact actual frame supplies a finite common-form certificate; numerical stabilizer resources require its actual coefficients.

For the retained receiving geometry there is a concrete specialization. The original receiver radius is below 65, the prescribed quarter-age gate ends before T_0, and $65 + n_+ t_* < 2 \cdot 10^6$. The retained compact auxiliary is identically physical on that entire backward cap worldtube. This is the causal-agreement statement used for the incoming homogeneous transport estimate, so its exterior does not alter the local field gate. Use this same fresh postdecoder stage, with its gate still before T_0; the later translation and hold do not need to fit inside that interval. The Chapter IV source-energy certificate has $Q_\tau \geq (7/8)X$, $X \geq H_f/2$, and physical lapse at least $n_- = 1997/2000$. After the retained scalar shift its coefficient is at least $7n_-/16 > 1/4$, so one may use $a_0 = 1/4$ with a strict margin. Its inverse-frame weighted operator bound is

$$\begin{gathered} \|\omega^{-1/2}(p \pm q - I)\omega^{1/2}\| \leq 401A, \\ A = 2^{-83}. \end{gathered} \tag{157}$$

For a real coordinate cap test the pulled smearing is $(p \pm q)f$. Consequently $\mu_* \leq (1 + 401A)^2 \mu_c < C_f^2 \mu_c$, and a sufficient choice is

$$s(t) = 8C_f^2 \mu_c h(t)^2. \tag{158}$$

This specialization uses the weighted inverse-frame bound, not an assertion of old-Fock form-domain preservation. Source-region agreement alone would be insufficient: the full detector-worldtube agreement is used here explicitly. A different geometry or longer field gate requires its own version of that causal and common-form certificate. Here $S_g = 8C_f^2\mu_c \int h(t)^2\,dt$, and the peak and total variation are the same coefficient times $\|h\|_\infty^2$ and $\mathrm{TV}(h^2)$, respectively.

Put $S_g = \int_{\mathrm{gate}} s(t)\,dt$. Since P^2 commutes with the field gate interaction at every time, this addition changes only a known free pointer shear. Its exact pointer equation is

$$Q_{\mathrm{gate}} = Q + 2(k\ell + S_g + c_h)P + X. \tag{159}$$

The initially prepared pointer is independent of the entire B–source–field state. This pointer preparation is the only product preparation used in the following Gaussian convolution; the bit and continuum need not be independent.

14.3 An exact finite controlled translation

Put $Z_B = |0\rangle\langle 0| - |1\rangle\langle 1|$. After the field gate has ended, assign a local pointer-coordinate control of duration $\tau > 0$,

$$H_w(t) = kP^2 + \lambda(t)Z_B P + \frac{\lambda(t)^2}{4k}I = k\left(P + \frac{\lambda(t)Z_B}{2k}\right)^2 \geq 0. \tag{160}$$

The real pulse has area $\int_0^\tau \lambda(t)\,dt = a > 0$. All displayed generators commute, so

$$U_w = \exp\left[-ik\tau P^2 - iaZ_B P - i\int_0^\tau \frac{\lambda(t)^2}{4k}\,dt\right]. \tag{161}$$

This is a direct sum of translated massive-pointer Hamiltonians on the two bit sectors. The unbounded momentum is handled by its spectral calculus and positive quadratic completion, rather than called a bounded bit control. On constant pulse segments the operators are self-adjoint on the momentum-squared domain; their form domain requires only finite $\langle P^2\rangle$. The exponential is defined on the whole Hilbert space. This adds a specified bit-to-meter interaction, not a derived microscopic SAM actuator or an autonomous controller.

Focus the initial Gaussian pointer for the known combined gate-and-write duration: set

$$\begin{gathered} Y = Q + 2[k(\ell + \tau) + S_g]P, \\ \langle Y\rangle = \langle P\rangle = 0, \\ \mathrm{Var}\,Y = \sigma^2, \quad \mathrm{Var}\,P = \frac{1}{4\sigma^2}, \\ \mathrm{Cov}_{\mathrm{sym}}(Y, P) = 0. \end{gathered} \tag{162}$$

After writing and a further free hold $s \geq 0$, the exact output coordinate in the initial picture is

$$Q_{\mathrm{out}}(s) = Y + X + 2(c_h + ks)P + aZ_B. \tag{163}$$

The write drift and the stabilizer's known shear have been included in the focus. Simply reusing a focus for the field gate alone would leave additional drift in the noise. The stabilizer leaves the actual X, c_h and source backreaction law unchanged. The bit commutes with the actual joint field gate and with X, even when its state is correlated with the field.

14.4 A finite retention window without an intrinsic mean floor

Declare one read at any chosen hold time $s \in [0, T_A]$. The background coordinate $Q_b(s) = Y + X + 2(c_h + ks)P$ has uncentered

second moment

$$\langle Q_b(s)^2\rangle \le M_{\rm rec} + \sigma^2 + \frac{(c_{\max}+kT_A)^2}{\sigma^2}. \tag{164}$$

The cross expectations between the centered initial Gaussian variables and X vanish by their initial product preparation. This statement does not factor an expectation involving Z_B and X. Choose the finite width

$$\sigma^2 = c_{\max} + kT_A > 0. \tag{165}$$

Then the following uniform bound holds:

$$\begin{aligned} B_A &= M_{\rm rec} + 2(c_{\max}+kT_A) \\ &= 6C_f^2\mu_c E_{\rm rec} + 4C_f^2 V_c \\ &\quad + 48g^2C_f^2 n_+^2 \mu_{\rm sh} t_*^2 D_c + 2kT_A, \\ \sup_{0\le s\le T_A} \langle Q_b(s)^2\rangle &\le B_A. \end{aligned} \tag{166}$$

The positive width follows from the nonzero cap test $V_c > 0$. Thus the construction gives a finite noisy position record for a declared finite hold interval. It does not assume an indefinitely stable free pointer or an unmodelled sequence of nondestructive reads.

Decode pointer positions $q > 0$ as bit zero and $q \le 0$ as bit one. Since Z_B and $Q_b(s)$ commute, their spectral probabilities are an ordinary joint probability distribution, with arbitrary correlations. If the pointer sign disagrees with the stored bit, then $|Q_b(s)| \ge a$. Spectral Markov therefore gives

$$P(\widehat{r} \ne r) \le \frac{\langle Q_b(s)^2\rangle}{a^2} \le \frac{B_A}{a^2}. \tag{167}$$

No conditional moment divided by the probability of a correct source bit is used. For each true source-age label j, if the source decoder has error at most $\varepsilon_{\rm src}^{\rm ref}$ on the reference input, the union of the source-error and transfer-error events gives

$$\varepsilon_{\rm final}^{\rm ref} \le \min\left\{1, \varepsilon_{\rm src}^{\rm ref} + \frac{B_A}{a^2}\right\}. \tag{168}$$

This inequality needs neither independence of errors nor a known mean of the actual field. Any desired positive transfer allowance η has the finite sufficient displacement $a \ge \sqrt{B_A/\eta}$.

For a subsequent smooth sign-controlled memory write, let its trusted positive and negative position bands start at w and $-w$, with $0 < w < a$. The probability of being in the wrong band or its unresolved central region obeys the same joint-event argument:

$$P(\text{position outside its bit's trusted band}) \le \frac{B_A}{(a-w)^2}. \tag{169}$$

This is a one-time position probability at the specified memory-entry time. Motion during a later memory gate requires that gate's own bound; it is not an assertion that the pointer is parked.

If the signed displacement magnitude differs from a by at most δ_a and the threshold has position error at most δ_q, replace the denominator in Eq. (167) by $(a-\delta_a-\delta_q)^2$, with a strictly positive margin. Nearest-bin rounding of width w can be included by increasing δ_q by $w/2$. This is a finite threshold margin, not a demand for an infinitely fine position ruler. A finite aperture $|q| \le L_q$ can retain overflow as an explicit failure outcome. Its additional probability is at most

$$P(\text{overflow}) \le \frac{(\sqrt{B_A}+a+\delta_a)^2}{L_q^2}. \tag{170}$$

The estimate uses the second-moment triangle inequality, with no factorization of bit and background. No overflow is silently discarded.

14.5 The complete receiving POVM and instrument

The actual Gaussian pointer variance at hold time s is $\nu(s) = \sigma^2 + (c_h + ks)^2/\sigma^2$. Let Γ_ν be its centered Gaussian probability measure and let E_X be the actual spectral measure of X. With $I_0 = (0,\infty)$, $I_1 = (-\infty, 0]$ and $\Pi_r = |r\rangle\langle r|_B$, the joint bit/output effects in the unmeasured source–field picture are

$$\begin{aligned} F_{r,y} &= \Pi_r \int \Gamma_{\nu(s)}\big(I_y - x - a(-1)^r\big)\, E_X(dx), \\ F_y &= \sum_r F_{r,y}. \end{aligned} \tag{171}$$

These positive effects sum to the identity. They are the actual Chapter IV Gaussian-convolution law with the added controlled displacement. They retain the state-dependent continuum spectral law; it has not been replaced by a fitted cap mean. Unknown c_h has its stated uniform bound, while its actual value defines the exact instrument.

For completeness let $U_{\rm all}(s)$ denote the actual nonuniform joint evolution, field gate, write unitary and free hold, including the unmeasured source–field drift. For the supplied independent Gaussian pointer state σ_D, define

$$\begin{aligned} \mathscr{I}_y(\rho) &= (I\otimes\mathscr{P}_y)U_{\rm all}(s)(\rho\otimes\sigma_D)U_{\rm all}(s)^*(I\otimes\mathscr{P}_y), \\ \mathscr{P}_y &= \mathbf{1}_{I_y}(Q). \end{aligned} \tag{172}$$

Here ρ includes the bit, source and full field, and the pointer is kept in the quantum output. The maps are completely positive and their sum is trace preserving. A classical outcome register may be added as $\sum_y |y\rangle\langle y| \otimes \mathscr{I}_y(\rho)$. The overflow refinement likewise includes every outcome. Tracing some recipients for a reported marginal does not remove them from the physical resource account.

Equation (172) is the ideal sharp position-event instrument. A discontinuous position projection generally does not preserve finite pointer kinetic energy, so it is not assigned a finite registration-work implementation. A smooth finite-energy target instrument can instead use real smooth multipliers

$$\begin{aligned} m_0(q) &= \cos\vartheta(q), \\ m_1(q) &= \sin\vartheta(q), \\ \vartheta(q) &= \begin{cases} \pi/2, & q \le -w, \\ 0, & q \ge w, \end{cases} \\ 0 &\le \vartheta \le \pi/2, \end{aligned} \tag{173}$$

with bounded derivatives and a smooth connection inside $(-w, w)$. Replace each $\mathscr{P}_y$ in the retained-pointer map by

$m_y(Q)$. The resulting maps again form a complete instrument. Their effects on the bit–source–field input are the convolution of m_y^2 with the same displaced actual continuum Gaussian law. Outside the central band they return the correct sign with certainty, so the bit-transfer error is bounded by Eq. (169). For any finite pointer momentum second moment, their nonselective output obeys

$$\begin{aligned}\sum_y \|Pm_y(Q)\psi\|^2 &= \|P\psi\|^2 + \langle\psi, \vartheta'(Q)^2\psi\rangle \\ &\le \|P\psi\|^2 + \|\vartheta'\|_\infty^2\|\psi\|^2.\end{aligned} \tag{174}$$

This identity extends to mixed and entangled inputs. It describes the real-multiplier target instrument. The positive memory pulse in the next section has the same effects but includes its coordinate-dependent common phase: with $\vartheta = \pi A_w/2$, its Kraus operators are $e^{-i\vartheta}\cos\vartheta$ and $ie^{-i\vartheta}\sin\vartheta$. For that instrument the corresponding identity is $\sum_y \|PK_y\psi\|^2 = \|(P-\vartheta')\psi\|^2 + \langle\vartheta'^2\rangle$. Its finite-write estimate below includes this phase through the full multiplier derivative bounds. A two-state register has the bounded frozen-coordinate target dilation generated by $\vartheta(Q)$ times its Pauli rotation. Its finite-duration physical implementation must retain the active kP^2 drift and charge the corresponding memory-gate error; the smooth target is not a hidden pointer-parking assumption.

If the original preparation is within δ_{prep} of the reference and the specified access, storage or registration channel has a further worst-label joint-state error budget δ_{access}, contractivity of this same complete channel gives

$$\varepsilon_{\text{final}} \le \min\left\{1, \varepsilon_{\text{src}}^{\text{ref}} + \frac{B_A}{a^2} + \delta_{\text{prep}} + \delta_{\text{access}}\right\}. \tag{175}$$

Use the calibrated denominator and overflow term when those outputs are part of the declared protocol. The preparation distance is paid once on the entire channel; it is not also included in $\varepsilon_{\text{src}}^{\text{ref}}$. If a previously quoted source ready-error budget already includes that distance, reassemble its stated additive components rather than adding the same preparation allowance twice. Ready failures remain explicit, and counting a per-label failure at most f as unsuccessful gives the all-attempt error $f + (1-f)\varepsilon_{\text{final}}$.

14.6 Finite energy, switching work and causal event map

The focused Gaussian has finite pointer momentum energy $k\langle P^2\rangle = k/(4\sigma^2)$ and finite initial position variance $\sigma^2 + [k(\ell+\tau)+S_g]^2/\sigma^2$. Momentum is conserved through the field gate, translation and hold. For a rectangular write with $\lambda = a/\tau$, the two jumps of Eq. (160) have total absolute mean work bounded by

$$W_{\text{write}}^{\text{abs}} \le 2|\lambda|\sqrt{\langle P^2\rangle} + \frac{\lambda^2}{2k} = \frac{a}{\tau\sigma} + \frac{a^2}{2k\tau^2}. \tag{176}$$

This includes the positive scalar completion on both jumps. The signed write payment telescopes for this commuting control; its absolute borrow, controller capital and pointer preparation are still charged. The actual field-gate switching, deposited energy and backreaction retain the Chapter IV ledger. The added stabilizer has finite peak expected capital $\|s\|_\infty/(4\sigma^2)$ and additional absolute work at most

$$W_{\text{stab}}^{\text{abs}} \le \langle P^2\rangle \int |s'(t)|\,dt = \frac{\text{TV}(s)}{4\sigma^2}, \tag{177}$$

including endpoint jumps if used. Its known focus preparation also has the finite position variance just displayed. No global renormalized vacuum-ramp work or thermal reset cost is inferred from Eq. (176).

One explicit finite model choice is

$$\begin{aligned} a &= 2^{10}, \\ k &= 2^{-40}, \\ \tau &= 2^{20}, \\ \lambda &= 2^{-10}, \\ T_A &= 2^{20}. \end{aligned} \tag{178}$$

Its pointer mass in the convention $k = 1/(2M_D)$ is 2^{39}, and its write completion height is $\lambda^2/(4k) = 2^{18}$. The focused variance is $\sigma^2 = c_{\max} + 2^{-20}$ and $B_A = B_{\text{N2}}(E_{\text{rec}}, t_*) + 2^{-19}$. With trusted-band width $w = a/2$, Eq. (169) is $4B_A/a^2$. These numbers exhibit finite amplitude, capital and latency in model units; they are not measured SI calibration. The field-gate stabilizer still pays its actual certified S_g, peak and variation separately.

The source decoder, actual continuum gate and added translation have separate finite durations. The retention window starts after writing; the receiving result exists only after the chosen hold and registered read. A spatially local Z_BP actuator requires the bit carrier to be co-located with the selected meter. If the decoded bit is delivered from elsewhere, its causal transport, delay and possible errors belong in the declared access map and the new t_* chronology. It is not supplied instantaneously to a remote detector. The same fixed schedule and classification rule apply to both unknown source-age hypotheses. The source-clock phase certificate is needed before the bit is decoded. The new t_* concerns the finite actual continuum stage through its field gate; the long write and hold occur after that coupling is off and do not extend a source-clock or field-gate certificate to later times.

This supplies a complete continuum-backed receiving instrument and a finite noisy pointer-retention model under its stated actuator and access laws. The intrinsic nonuniform capmean transfer, microscopic actuator, achieved calibration and a physical carrier's durable-memory/access performance remain distinct operands; none is replaced by the added known displacement.

15. A finite pointer-to-memory instrument with causal local access

Source and scope. The retained SAM_land donor readout construction supplies neutral record bits with degenerate rest channels and positive bounded couplings proportional to $I - X$; its finite mark/write/unmark law retains the complete complementary cloud. Its final record is held, with reset treated separately (`samland_donor_readout1/DERIVATION.md`, Section 4, and its `ENERGY_ACCOUNT.md`). We reuse that kind of record recipient below, with a new, explicitly specified smooth coordinate response. The physical memory covariance of `volii_physical_memory1` and the colored classical

field bath of `voli_radiative_memory1` retain their own source laws; neither supplies a spin-register flip rate here.

The new instrument, finite-write estimate and correlated-hold bounds are developed in campaign voli_chvi_detector_record1_20261017, with four native SLC-GEN5-R1 calls. They join a decoded bit to the actual detector pointer under the stated receiving law. The coordinate-to-qubit coupling specified below is an explicit apparatus assignment; it is not derived here from a microscopic material actuator. They do not extract a new positive cap mean from the old field, assign a microscopic material or temperature, or replace the later two-arm transport calculation. Local controls in this section require the decoded bit, detector pointer and memory to be physically co-located, or a previously specified causal delivery channel bringing the bit to that location.

15.1 A correlated loaded pointer and its complete binary conversion

Let B be the isolated decoder-output qubit from the source-only protocol, with $P_b = |b\rangle\langle b|$, $b \in \{0,1\}$. For each operational input label j, let ρ_j be the finite-energy reference output of the actual decoder pulses on the ideal prepared marker. Its error obeys $\mathrm{Tr}(P_{1-j}\rho_j) \le e_{\mathrm{dec,ref},j}$, excluding the marker preparation distance δ_{prep}. The field can remain correlated with this bit. Let the actual analog detector noise observable be N, acting on the pointer/source/field factors outside B, so $[N,P_b]=0$. Assume the retained full detector calculation supplies the per-label moment bound

$$\begin{gathered}\mathrm{Tr}(\rho_j N^2) \le \mathscr{B}_{\mathrm{ret}} \\ \text{uniformly over the declared} \\ \text{analog receiving interval.}\end{gathered} \tag{179}$$

The joint state ρ_j may correlate B, N and all other degrees of freedom. No conditioning on an exceptionally rare bit value is used to claim the same moment bound.

After the detector field gate, the specified bit-controlled translations load the same pointer with signs $+a$ for $b=0$ and $-a$ for $b=1$. Its loaded coordinate is therefore

$$\begin{gathered}Q = N + a(P_0 - P_1), \\ a > 0.\end{gathered} \tag{180}$$

Equivalently, in a Schrödinger description the loading unitary translates the ordinary pointer coordinate and the displayed equation specifies its input Heisenberg observable. The source-defined field distribution and its correlations remain in N. The added sign is the decoded bit's paid apparatus displacement, not an assumed intrinsic mean of N.

Choose $0 < w < a$. With $x = (q+w)/(2w)$ define the C^1 threshold response

$$\begin{gathered}A_w(q) = \begin{cases} 1, & q \le -w, \\ 1 - 3x^2 + 2x^3, & -w < q < w, \\ 0, & q \ge w. \end{cases} \\ 0 \le A_w \le 1, \\ G := \|A_w'\|_\infty = \frac{3}{4w}.\end{gathered} \tag{181}$$

This A_w is an apparatus response, not the accumulation scalar $A(r)$ or the computational `A_OPERATOR`.

For a finite registration aperture, take $L_q > w$ and a C^1 multiplier χ_{L_q} equal to one on $|q| \le L_q$ and zero on $|q| \ge 2L_q$, with cubic shoulders and $\|\chi_{L_q}'\|_\infty \le 3/(2L_q)$. Use $A_{w,L_q} = \chi_{L_q} A_w$ in the physical memory pulse. Its response remains in $[0,1]$ and has compact coordinate support. The two derivative supports are disjoint, so $\|A_{w,L_q}'\|_\infty \le \max\{G, 3/(2L_q)\}$. The finite-pulse derivative bounds below apply with that derivative bound. The uncut threshold formulas acquire the following additional wrong-bit probability, charged once through the final aperture/read allowance:

$$\varepsilon_{\mathrm{ap}} \le \frac{(\sqrt{\mathscr{B}_{\mathrm{ret}}} + a)^2}{L_q^2} \le \frac{2(\mathscr{B}_{\mathrm{ret}} + a^2)}{L_q^2}. \tag{182}$$

This follows from $Q = N + aZ_B$ and the second-moment triangle inequality. The trusted-band bound is supplemented by this aperture term; its output is still part of the complete binary experiment. For $L_q = 2^{20}$ and $w = 2^9$, the original G is unchanged. The final budget can absorb this probability into a stated combined aperture/read allowance.

Supply one memory qubit R, initially $|0\rangle_R$, with degenerate rest Hamiltonian, and set $X_- = (I - X_R)/2$ and $D_w = A_w(Q) \otimes X_-$. The ideal positive write pulse has area π:

$$\begin{gathered}V_\theta = e^{-i\theta D_w}, \quad 0 \le \theta \le \pi, \\ U_w = V_\pi.\end{gathered} \tag{183}$$

It is exactly the identity on the memory for $q \ge w$, and exactly X_R for $q \le -w$. At every coordinate it is unitary, including the finite transition band. Reading the memory in its Z_R basis gives the two Kraus operators on the retained incoming system

$$\begin{aligned}K_0(Q) &= e^{-i\pi A_w(Q)/2}\cos[\pi A_w(Q)/2], \\ K_1(Q) &= ie^{-i\pi A_w(Q)/2}\sin[\pi A_w(Q)/2],\end{aligned} \tag{184}$$

with $K_0^* K_0 + K_1^* K_1 = I$. Both outcomes and their conditional residual source/pointer/field states are retained. The transition band is not postselected away; its possibly erroneous binary output is included. The memory need not be read immediately: Eq. (183) first creates a coherent memory correlation, to which the hold channel below can be applied.

Proposition (threshold error without independence). For each label j, the ideal write obeys

$$e_{\mathrm{thresh,ref},j} \le \min\left\{1, e_{\mathrm{dec,ref},j} + \frac{\mathscr{B}_{\mathrm{ret}}}{(a-w)^2}\right\}. \tag{185}$$

Proof. On the simultaneous spectral subspace where the decoded bit equals j and $|N| < a - w$, Eq. (180) lies strictly beyond the correct side of the transition band. The wrong memory effect vanishes there. Since it is a positive contraction, the commuting spectral calculus gives

$$K_{1-j}(Q)^* K_{1-j}(Q) \le P_{1-j} + \mathbf{1}_{\{|N| \ge a-w\}}. \tag{186}$$

The spectral second-moment inequality bounds the second term's expectation by $\mathscr{B}_{\mathrm{ret}}/(a-w)^2$. Taking the expectation in the full joint state establishes Eq. (185), with no factorization of bit and detector noise. □

15.2 A finite positive pulse with the pointer kinetic energy retained

Let the pointer's actual rest Hamiltonian be $H_P = kP^2$, $k > 0$, $P = -i\partial_Q$, and retain it throughout the write. A rectangular pulse of duration $\tau_w > 0$ is

$$H_{\text{write}} = kP^2 + \frac{\pi}{\tau_w} D_w, \qquad U_{\text{write}} = e^{-i\tau_w H_{\text{write}}}. \tag{187}$$

Both terms are positive; there is no kinetic parking. The actual source–field generator $H_{\text{sf}}(t)$ is a spectator commuting with these pointer and memory operators. It factors from the comparison even when its state is entangled with the pointer. Its own evolution and energy are retained. Equation (187) specifies the write-stage memory law: any additional noncommuting bath coupling during that stage needs a separate write-error bound. Activating a different hold coupling afterward retains its switching work rather than parking the pointer.

Suppose the joint write-entry state has $\langle P^2 \rangle \le p_2$. In the intended pipeline the detector gate and bit-controlled translations depend on the analog pointer momentum and commute with P^2; its free analog hold does also. Thus the declared Gaussian input moment $p_2 = 1/(4\sigma^2)$ travels unchanged through those stages. A further control that does not preserve this moment needs its own bound. Purification includes arbitrary correlations with the remaining systems. The bounded C^1 multiplier A_w preserves $D(P)$, and for a normalized purification ψ,

$$\|PV_\theta \psi\| \le \sqrt{p_2} + |\theta| G, \qquad \|PD_w V_\theta \psi\| \le \sqrt{p_2} + (|\theta| + 1) G. \tag{188}$$

These follow from $[P, D_w] = -iA_w'(Q)X_-$ and differentiation of the commuting multiplication exponential. In particular both ideal orbits needed by the bounded-control comparison have free-energy form expectation at most

$$C_w = k[\sqrt{p_2} + (\pi + 1)G]^2 \le C_w^{\text{up}} := 2k[p_2 + (\pi_* + 1)^2 G^2], \qquad \pi_* \ge \pi. \tag{189}$$

No $D(P^2)$ assumption on the initial state is required for this estimate. The self-adjoint Hamiltonian in Eq. (187) is the nonnegative free Hamiltonian plus a bounded self-adjoint control.

Proposition (finite-write discrepancy). For the specified reachable input family,

$$\|(U_{\text{write}} - U_w)\psi\| \le r_w := (1 + 2\pi)\sqrt{2\tau_w C_w} \le (1 + 2\pi_*)\sqrt{2\tau_w C_w^{\text{up}}}. \tag{190}$$

The corresponding complete joint-output trace distance and each memory-outcome probability error are at most $\min\{1, r_w\}$.

Proof. Set $F(t) = e^{-itH_P}$ and move the actual pulse into its free interaction picture. Its bounded generator is $(\pi/\tau_w)F(t)^* D_w F(t)$; the ideal one is $(\pi/\tau_w)D_w$. For any form-domain vector ξ, the spectral theorem gives

$$\|(F(t) - I)\xi\|^2 \le 2|t| \langle \xi, H_P \xi \rangle. \tag{191}$$

Expanding $F(t)^* D_w F(t) - D_w = F(t)^* D_w (F(t) - I) + (F(t)^* - I) D_w$ and using Eq. (189) bounds its action on the ideal orbit by $2\sqrt{2tC_w}$. Bounded-generator Duhamel therefore bounds the interaction-picture difference by $2\pi\sqrt{2\tau_w C_w}$, using $t \le \tau_w$ under the integral. Returning from that picture costs at most $\sqrt{2\tau_w C_w}$ on the final ideal orbit. Add the two bounds. Purifying a mixed input and then taking partial traces gives the same output trace-distance conclusion. □

15.3 Passive hold and correlated transverse disturbances

After writing, turn off the write pulse. The memory rest law is the same degenerate channel used by the retained SAM_land record construction. A more general admitted passive hold is

$$H_d(t) = |0\rangle\langle 0|_R \otimes L_0(t) + |1\rangle\langle 1|_R \otimes L_1(t), \qquad [H_d(t), Z_R] = 0, \tag{192}$$

with well-defined block propagators on the retained environment. This environment includes the pointer, source, field and every noise reference needed to describe correlations. The memory may dephase or acquire joint environmental records, but each Z_R population is exactly constant. Thus its readable binary record is held for every declared finite interval under this law. No refresh or measurement is needed during the hold.

For an actual transverse disturbance, split any admitted additional interaction into its diagonal part, absorbed in H_d, and its bounded off-diagonal part

$$V_{\text{off}}(t) = \Pi_0 V(t) \Pi_1 + \Pi_1 V(t) \Pi_0, \quad \Pi_b = |b\rangle\langle b|_R \otimes I, \qquad \Lambda_T = \int_0^T \|V_{\text{off}}(u)\| \, \mathrm{d}u. \tag{193}$$

The norm is on the full retained joint system, not an average of independent cell errors. Interaction-picture bounded-perturbation Duhamel gives a joint-channel trace-distance error at most $\min\{1, \Lambda_T\}$ on every input and reference. Consequently the wrong-label population can increase by at most $\min\{1, \Lambda_T\}$ uniformly for $0 \le t \le T$.

A sharper population envelope is also available. If its initial value is e_0, then

$$e(t) \le \sin^2\left(\min\left\{\frac{\pi}{2}, \arcsin\sqrt{e_0} + \Lambda_t\right\}\right). \tag{194}$$

To see this, purify the complete state. In the diagonal interaction picture, the wrong-sector probability obeys, almost everywhere, $|\dot{e}| \le 2\|V_{\text{off}}\| \sqrt{e(1-e)}$ by Cauchy–Schwarz between the two orthogonal sectors. Integrating the bound on $\arcsin\sqrt{e}$ gives Eq. (194), with endpoint values obtained by continuity. For $e_0 = 0$ this becomes $e(t) \le \sin^2(\min\{\pi/2, \Lambda_t\})$. Use this population envelope or the additive channel error in a budget; do not charge the same disturbance twice.

This is a passive finite-spin receiving model, not an assigned Markov flip rate. For example a finite correlated thermal environment may be declared by a positive Ising Hamiltonian

$$H_E = \sum_\ell h_\ell \frac{I - Z_\ell}{2} + \sum_\ell J_\ell \frac{I - Z_\ell Z_{\ell+1}}{2}, \qquad h_\ell, J_\ell \ge 0, \quad \rho_E = \frac{e^{-\beta H_E}}{\operatorname{Tr} e^{-\beta H_E}}, \tag{195}$$

and a static coupling $V_{\mathrm{off}} = X_R \otimes B_x + Y_R \otimes B_y$. Its actual matrices give $\Lambda_T \le T(\|B_x\| + \|B_y\|)$, regardless of thermal correlations. Diagonal Z_R couplings may be retained in Eq. (192). If a nonnegative finite total bath Hamiltonian is desired, include the common offset $\|V_{\mathrm{off}}\| I$ (and the norm of any additional bounded diagonal coupling) in its energy account. This does not change the memory probabilities. The bath preparation, β, couplings and energy offsets would be new receiving operands; no such numerical thermal model is inferred from an isotope covariance or inserted into the example below. No thermal activation advantage is claimed by this state-independent norm bound.

15.4 Readout, complete outcomes and causal access

For local readout supply a blank qubit A. A positive bounded pulse

$$\begin{aligned} H_{\mathrm{copy}}(t) &= u(t)\Pi_1 \otimes X_{-,A}, \\ u(t) \ge 0, &\quad \int u(t)\,\mathrm{d}t = \pi \end{aligned} \tag{196}$$

copies the memory's orthogonal Z_R label to A. It commutes with the diagonal hold law, even with the full correlated environment retained. The final Z_A measurement uses the retained Born/instrument premise. This specifies its ideal back-action: it preserves memory label populations, while permitting correlation and dephasing. An actual copy/read channel may instead carry a declared outcome error δ_{read} uniform over the receiving interval. A transverse exposure during copying is included either in the extended hold exposure or in that read error, once.

More explicitly, first append the blank memory, apply the finite write, then the full hold channel, append the read qubit and apply its copy unitary. Projection onto $|b\rangle_A$ followed by the read-register partial trace defines a completely positive instrument map $\mathscr{J}_b^t$ on the retained systems. The two maps sum to a trace-preserving channel. No residual source, pointer or environmental cloud is replaced by a pure state merely because a classical outcome has been registered. For an input already carrying the chapter's ready/failure flags, define

$$\begin{aligned} \mathscr{I}_b^t(\widehat{\rho}) &= \mathscr{J}_b^t(P_{\mathrm{ready}}\widehat{\rho}P_{\mathrm{ready}}), \\ \mathscr{I}_{\mathrm{fail}}^t(\widehat{\rho}) &= P_{\mathrm{fail}}\widehat{\rho}P_{\mathrm{fail}}, \end{aligned} \tag{197}$$

with multiple failure flags kept as separate blocks when supplied. Their traces sum to one. Failure is a recorded outcome, not a missing attempt.

For the closed local protocol let the receiving region contain the memory worldtube and read apparatus. Its access channel is the local channel just specified, available after the finite write/copy completion. Retention is tested for one read at any chosen time in its declared interval. Repeated reads require their additional copy gates, blank qubits, disturbance and work accounts; single-read performance does not automatically pay for them.

For a remote region, an additional causal delivery channel is an explicit operand. Before its first allowed arrival t_{arr} the receiver output has a common no-access flag, described by the trace-preserving replacement map $\omega \mapsto \mathrm{Tr}(\omega)|\perp\rangle\langle\perp|$. After arrival, a supplied delivery channel and its error can be composed with the memory readout. The actual Chapter IV hyperbolic propagation constrains source-generated changes outside their causal future; neither this memory coupling nor the word "access" supplies an instantaneous remote control. The local construction is complete without selecting a new remote carrier. Its remote extension must supply arrival and delivery accuracy, rather than infer them from a finite register circuit. Memory survival on $[0, T_{\mathrm{store}}]$ and remote readability after t_{arr} are therefore distinct intervals.

15.5 The resulting uniform error and resource ledger

Let $\delta_{\mathrm{acc}} = \delta_{\mathrm{read}} + \varepsilon_{\mathrm{ap}}$ include the physical aperture term when that response is used. Let δ_{init} bound any additional joint ready-state error for the analog-pointer and blank-memory receiving preparation, relative to the family for which the moment and momentum bounds above hold. Propagate this error as a trace distance through the complete instrument. It does not imply a second-moment bound for an arbitrary nearby actual state, and it is not a second charge for marker preparation. First apply the threshold and finite-write estimates to the finite-energy reference pipeline. The actual marker preparation is within δ_{prep} of that reference at the original input; contraction through the entire actual receiving channel adds δ_{prep} once at its final output. For compact notation write $e^{\mathrm{r}}_{\mathrm{dec}} = e^{\mathrm{r}}_{\mathrm{dec,ref}} + \delta_{\mathrm{prep}}$, as in the inherited decoder budget. This notation includes the final whole-channel preparation payment; it does not transfer a moment bound to an arbitrary nearby actual state. Combining the preceding estimates gives the ready-conditional local bound

$$\boxed{e^{\mathrm{r}}_{\mathrm{rec}} \le \min\left\{1, e^{\mathrm{r}}_{\mathrm{dec}} + \frac{\mathscr{B}_{\mathrm{ret}}}{(a-w)^2} + r_w + \delta_{\mathrm{init}} + \Lambda_{T_{\mathrm{store}}} + \delta_{\mathrm{acc}}\right\}.} \tag{198}$$

All terms are per-label uniform bounds on the states where they are used. The threshold argument handles bit/field correlations directly, and the channel/hold arguments retain their environment. No independence between decoding, analog noise, writing, storage and access is required. For a complete attempt with failure probability at most f, counted as unsuccessful, the bound becomes

$$e^{\mathrm{all}}_{\mathrm{rec}} \le f + (1-f)e^{\mathrm{r}}_{\mathrm{rec}}. \tag{199}$$

If failure was already included in an all-attempt channel error, it must not be charged again by Eq. (199).

A rectangular write has peak π/τ_w, operator-norm action π, and switch-work variation at most $2\pi/\tau_w$ in the common model chart. Indeed its turn-on work lies in $[0, \pi/\tau_w]$ and its turn-off work in $[-\pi/\tau_w, 0]$. For finite-rest-energy inputs the signed total equals the change of pointer rest energy when the controller is zero at both endpoints; the spectator source–field energy is conserved by this coupling. This signed work is not identified with heat. The finite copy pulse has the same action and, at duration τ_r, peak π/τ_r and switch-work variation at most $2\pi/\tau_r$. Its ideal diagonal-hold commuting limit has cancelling signed on/off work; peak and temporary borrow remain charged.

The ledger additionally retains the upstream field-gate and bit-loading translation controls, the pointer kinetic preparation, one fresh blank memory bit and one read bit per supplied protocol, all failed preparation attempts and their ready/failure

preparation accounts, any hold bath and coupling energy, physical delivery when used, and later reset/disposal. A degenerate rest energy is not a claim of free pure-state preparation or a Landauer heat formula. A cyclic erasure bound would require its own reservoir, temperature and logical ensemble. Model energies/work convert through the already declared action/time units; no SI calibration is inferred.

For the proposed arithmetic recipient one may use

$$\begin{aligned} a = 2^{10}, \quad w = a/2 = 2^9, \quad k = 2^{-40}, \\ T_{\mathrm{analog}} = 2^{20}, \quad \tau_w = 2^{-20}, \\ \sigma^2 = C_{\mathrm{prop}}^2 V_{\mathrm{up}}/2 + kT_{\mathrm{analog}}, \\ p_2 = \frac{1}{4\sigma^2}, \\ \mathscr{B}_{\mathrm{ret}} = \mathscr{B}_{\mathrm{new}} + 2kT_{\mathrm{analog}}. \end{aligned} \tag{200}$$

Here the propagation and detector ceilings retain their previously defined source meanings; these equations specify the receiving pointer rather than a measured calibration. The analog contribution is $4\mathscr{B}_{\mathrm{ret}}/a^2$, and the finite-write allowance $r_w \le 10^{-4}$ is used only after checking Eq. (190) with the corresponding outward inputs. A logical hold interval $T_{\mathrm{store}} = 2^{20}$ with $\Lambda_{T_{\mathrm{store}}} \le 10^{-4}$, together with $\delta_{\mathrm{init}} \le 10^{-4}$ and $\delta_{\mathrm{acc}} \le 10^{-4}$, is an explicit receiving specification, not an observed thermal lifetime. The exact local copy/read reference can take $\delta_{\mathrm{read}} = 0$; a finite aperture still retains its separately bounded tail. For the example, $\varepsilon_{\mathrm{ap}} \le 2 \cdot 10^{-6}$ and $\delta_{\mathrm{read}} \le 98 \cdot 10^{-6}$ fit the combined $\delta_{\mathrm{acc}} \le 10^{-4}$ allowance. The diagonal passive hold itself has $\Lambda_T = 0$; a nonzero exposure budget must be established from the actual admitted transverse interaction. These distinctions keep a constructive finite record law, its mathematical robustness, and achieved apparatus calibration as separately identified parts of the same end-to-end account.

16. Six-lane receiving values and a complete finite example

The native calculation first consumes the source decoder's actual form-energy and switching-work returns. It uses

$$\begin{aligned} E_{\mathrm{rec}}^{\mathrm{up}} = 2(C_{\mathrm{dec}}^{\mathrm{up}} + W_{\mathrm{dec}}^{\mathrm{up}}) + 2c^{\mathrm{up}}, \\ E_{\mathrm{rec}}^{\mathrm{up}} < 514719 \end{aligned} \tag{201}$$

in every lane, instead of importing the old accepted-state noise values. The cap operands are the retained upward dyadic enclosures

$$\begin{aligned} C_f &= \frac{100301}{100000}, & \mu_c^{\mathrm{up}} &= \frac{1310994537}{281474976710656}, \\ V_c^{\mathrm{up}} &= \frac{2553359}{70368744177664}, & D_c^{\mathrm{up}} &= \frac{79569}{281474976710656}. \end{aligned} \tag{202}$$

Each original signed coupling is bounded through its retained absolute upper endpoint in the second moment. The source-shell norm and actual elapsed horizon are retained from the same Chapter IV input. The finite-energy estimate applies to the unselected reference output of the actual decoder; preparation error is propagated once through the entire final channel.

Use the explicit pointer, translation and memory parameters

$$\begin{aligned} a = 1024, \quad w = 512, \quad L_q = 2^{20}, \quad k = 2^{-40}, \\ \tau_{\mathrm{tr}} = T_{\mathrm{analog}} = T_{\mathrm{store}} = 2^{20}, \qquad \tau_w = 2^{-20}, \\ \sigma^2 = \frac{C_f^2 V_c^{\mathrm{up}}}{2} + kT_{\mathrm{analog}}, \qquad p_2 = \frac{1}{4\sigma^2} \le 512^2. \end{aligned} \tag{203}$$

The focused Gaussian has $\sigma^2 < 9.7192641 \times 10^{-7}$ and pointer kinetic energy $kp_2 < 2.3394125 \times 10^{-7}$. The actual continuum/analog moment has $B_A < 16$ in all six lanes. Consequently the finite registration aperture contributes

$$\varepsilon_{\mathrm{ap}} \le \frac{2(B_A + a^2)}{L_q^2} < 2 \times 10^{-6}. \tag{204}$$

Its compact response has the same derivative bound $G = 3/(4w)$; the finite memory-pulse calculation is unchanged. The exact reciprocal candidate for p_2 is certified by a native product equal to one, before its use in that calculation.

The finite write discrepancy obeys

$$\begin{aligned} r_w^2 \le 4.7337967 \times 10^{-11}, \\ r_w \le 6.8802592 \times 10^{-6} < 10^{-4}. \end{aligned} \tag{205}$$

For a transparent composable budget retain the parent's declared age-phase 0.01, marker preparation distance 10^{-4} and source-control allowance 10^{-3}. Add receiving initialization, write and transverse-hold allowances each at most 10^{-4}. Allocate the final 10^{-4} access allowance as aperture at most 2×10^{-6} plus local copy/read error at most 98×10^{-6}. The resulting conservative bound is

$$\begin{aligned} e_{\mathrm{rec}}^{\mathrm{r}} \le e_{\mathrm{dec}}^{\mathrm{r}} + \frac{4B_A}{a^2} + \frac{3}{10000} + \frac{98}{10^6} + \frac{2(B_A + a^2)}{L_q^2} \\ \le 0.0015821358 \quad \text{in every original lane.} \end{aligned} \tag{206}$$

The bound uses the 10^{-4} write allowance, retaining the smaller calculated r_w as available margin. It does not silently reassign that margin to another control. All-attempt failure remains $f + (1-f)e_{\mathrm{rec}}^{\mathrm{r}}$. The bit and detector may remain correlated; no independence assumption was needed for these additions.

Resources and completion time. The pointer mass is 2^{39} in the convention $k = 1/(2M_D)$. The translation has $\lambda = 2^{-10}$, positive completion height 2^{18} and absolute switch-work bound 524289. The memory write has peak below 3294198.7 and absolute switch-work bound below 6588397.4. These finite quantities are model-unit payments. The detector stabilizer additionally charges its actual gate profile, quadratic capital and variation; the focused pointer's initial coordinate variance includes the known integrated stabilizer shear. All source even/-counterterm energy, original field-gate work and residual field states remain recipients.

A local read copy can have any declared positive finite duration and area π, with its finite peak and switch work as above. The record becomes readable after the source decode, continuum gate, translation, chosen analog wait, memory write, chosen logical hold and copy/read have finished. The source clock

Lane	Actual detector B_{new} upper	Added tail error upper	Full ready error upper
01	14.471425	0.000055204110	0.0015801122
02	14.473466	0.000055211897	0.0015801222
03	14.883113	0.000056774575	0.0015821358
12	14.471518	0.000055204465	0.0015801127
13	14.851770	0.000056655013	0.0015819817
23	14.819530	0.000056532025	0.0015818233

Table 3: Outward six-lane bounds for the declared receiving example. The tail column charges the pointer's central transition band. The full ready column also charges source decoding/preparation, finite memory writing, initialization, hold and the combined aperture/read allowance. It is uniform over the stated retention interval.

certificate refers to the earlier source entry. The long receiving latency is explicitly retained. Repeated reads, cyclic reset, bath preparation and remote delivery use additional supplied resources and their own errors.

17. Accessibility is a receiving operation

The domain in the phrase "accessible record" must be part of the experiment. A record may be perfectly distinguishable jointly and unreadable from a restricted subsystem. A signal may already have been emitted but not yet have reached a receiver. A memory may survive while its decoder is unavailable. These cases have different operational meanings.

For any two candidate memory states, a trace-preserving physical access channel $\mathscr{A}$ cannot increase their trace distance:

$$\tfrac{1}{2}\left\|\mathscr{A}(\rho_0)-\mathscr{A}(\rho_1)\right\|_1 \le \tfrac{1}{2}\left\|\rho_0-\rho_1\right\|_1 . \tag{207}$$

The proof is the variational characterization of trace distance: every allowed output POVM pulls back through the unital dual channel to an allowed input POVM. An inaccessible fragment can reduce readout performance; calling it inaccessible does not remove its contribution to the global state or restore a lost marginal fringe.

Chapter V adds a causal constraint to this account. An exterior record must reach an exterior receiver through an allowed physical process. Late receipt of signals from a finite source-age interval does not give the emitter an infinite history. A boundary convention that excludes a completed parent-ledger state supplies neither a quantum channel across that boundary nor a rule selecting one outcome. The state, field dynamics and receiving map remain necessary.

The finite tensor-product examples in this chapter are operational models of separated systems. They do not establish an exact factorization of a gravitational or constrained gauge-field Hilbert space across a horizon. If a proposed theory replaces that subsystem structure by an observable algebra, its accessible operations and probabilities must be specified there instead.

These restrictions preserve the source's physical distinction: a write is more than an unresolved possibility. They also prevent the word "write" from performing work that belongs to a Hamiltonian, an instrument, a decoder or a causal signal.

18. Two prescribed arms in one original continuum field

The following receiving construction replaces the isolated-arm substitution by a specified moving-source action. The geometry is a flat laboratory chart with field wave speed one in the original model units. Each rigid source centre follows a prescribed piecewise smooth timelike worldtube $x_j(t)$, with $v_j=\dot{x}_j$, $|v_j|\le 1/4$, and $n_j=\sqrt{1-|v_j|^2}$. Smooth pieces have two continuous derivatives in v_j,n_j; finitely many joins have continuous velocities. Splitting, trajectory control, and recombination are prescribed receiving operations. They are not newly derived gravitational or autonomous actuator laws. The intrinsic source is the original linear-current H_0 in Eq. (58); the distinct Chapter VII quadratic completion candidates are not silently substituted for that source.

18.1 A translated source action and the full field complement

Use one full one-particle space containing the two physical polarizations and every angular mode. Embed the original two normalized quadrupole patterns as orthonormal vectors $e_+,e_\times$. Their radial factor is the original u_R, so $u_a(\omega,\widehat{k})=u_R(\omega)e_a(\widehat{k})$ and $\|u_a/\sqrt{\omega}\|^2=\mu=1/(4R)$. An explicit fixed-axis realization uses the two unit Frobenius-norm trace-free quadrupoles $A_+=\mathrm{diag}(1,-1,0)/\sqrt{2}$ and $(A_\times)_{xy}=(A_\times)_{yx}=1/\sqrt{2}$. With $P=I-\widehat{k}\widehat{k}^T$, put $e_a=\sqrt{5/(8\pi)}[PA_aP-\frac{1}{2}P\,\mathrm{tr}(PA_a)]$. The spherical second/fourth moments give $\int e_a:e_b\,d\Omega=\delta_{ab}$. This fixes the actual angular profile and orientation of this receiving construction. Other orientations are simultaneous tensor rotations; no new amplitude is fitted. All orthogonal angular modes remain in the common Fock space, initially in their retained vacuum. Translation mixes these modes; none is dropped. Let

$$H_f=d\Gamma(\omega),$$
$$P_f=d\Gamma(\omega\widehat{k}),$$
$$T_x=e^{-ix\cdot P_f},$$
$$u_{a,x}(k)=e^{-ik\cdot x}u_a(k).$$

The interaction prescription is the original current paired with the field on that moving worldtube, integrated over its proper

time. Explicitly the boundary part of its action is

$$I_* \int dt\, n_j(t) \left[\varepsilon \sum_a J_a \phi(u_{a,x_j}) - h_s - H_{ct} \right], \quad J_+ = D,\ J_\times = C,$$

where $h_s = \varepsilon H_{\mathrm{pair}}$ and $H_{ct} = \varepsilon^2 \mu (D^2 + C^2)$. The common bulk field action uses laboratory time. The normalization is a declared rigid-quadrupole receiving incidence; boost deformations of the source profile and microscopic routing devices are additional physical operands. This action, rather than an independent bath for each arm, yields

$$H_j^{lab}(t) = H_f + n_j(t) T_{x_j(t)} (H_0 - H_f) T_{x_j(t)}^*. \tag{208}$$

Both original counterterms and all even energy are present. Scaling the interaction by n_j does not authorize replacing the retained source counterterm by a newly fitted static completion.

Put $\psi_j = T_{x_j}^* \Psi_j^{lab}$. Direct differentiation on a regularized field core, then closed-form extension, gives

$$\begin{aligned} K_j(t) &= H_f - v_j(t) \cdot P_f + n_j(t)(H_0 - H_f), \\ U_j^{lab}(t,s) &= T_{x_j(t)} U_{K_j}(t,s) T_{x_j(s)}^*. \end{aligned} \tag{209}$$

In particular the field momentum term is retained. Differentiating the singular translated form factor directly would need an ultraviolet norm not supplied by the original preparation. Equation (209) avoids that operation and preserves the original singular coupling. Because $|\langle \xi, P_f \zeta \rangle| \le \|H_f^{1/2} \xi\| \|H_f^{1/2} \zeta\|$, this new term is a continuous form on the original common energy domain.

18.2 Existence, energy, and a quantitative proper-age comparison

For a nominated original lane write $H_0 = \varepsilon e I + \varepsilon S + H_f + V + cI$, $c = 2g^2\mu$, preserving the sign of g in V. The retained annihilation-map estimate gives

$$|\langle V \rangle| \le 2\sqrt{c \langle H_f \rangle} \le \tfrac{1}{4} \langle H_f \rangle + 4c.$$

Define the positive comparison form

$$\begin{aligned} A_j &= K_j - n_j \varepsilon e I + \varepsilon I + 3cI, \\ b &= \varepsilon + 5c, \\ E_j(t) &= \langle A_j + bI \rangle. \end{aligned}$$

Then $A_j \ge H_f/2$, and its form norm is uniformly equivalent, after a positive shift, to $Q_f = D(H_f^{1/2})$. Its coefficients are piecewise C^2 as forms on this fixed domain. The constant-form-domain propagator theorem therefore supplies unique unitary weak dynamics, energy-domain propagation, and the form energy identity on each smooth piece; concatenate at the finite joins. The same construction on the finite direct sum of every source lane and four diagonal states gives the full source16 propagator. This application uses the common-domain hypotheses of the standard theorem explicitly.[29]

For normalized states the derivative obeys

$$\begin{aligned} |\dot{E}_j| &\le (2|\dot{v}_j| + |\dot{n}_j|) E_j, \\ E_j(t) &\le E_j(0) e^{B_j(t)}, \\ B_j(t) &= \int_0^t (2|\dot{v}_j| + |\dot{n}_j|) ds. \end{aligned}$$

Indeed $\langle H_f \rangle \le 2\langle A_j \rangle$ and $|\langle \varepsilon S + V + c \rangle| \le \langle A_j \rangle/2 + b$, which gives this inequality without requiring $\psi \in D(H_f)$. If the source enters at rest with the retained decoder form ceiling C, then $E_j(0) \le E_* = C + \varepsilon + 8c$.

The proper-age reference has generator $n_j H_0$, age $\eta_j(t) = \int_0^t n_j(s) ds$, and field energy at most $2(C+c)$ throughout. Its difference from the actual moving-frame generator is exactly

$$K_j - n_j H_0 = (1 - n_j) H_f - v_j \cdot P_f.$$

Weak Duhamel between the two energy-domain vectors gives

$$\begin{aligned} &\| [U_{K_j}(t,0) - e^{-iH_0 \eta_j(t)}] \psi \|^2 \\ &\le 2 \int_0^t \kappa_j(s) \sqrt{F_j(s) F_{ref,j}(s)}\, ds, \\ &\kappa_j = |1 - n_j| + |v_j|. \end{aligned} \tag{210}$$

Here $F = \langle H_f \rangle$. To see the norm estimate, differentiate the cross inner product as a Q_f, Q_f^* pairing. The difference form is bounded by $\kappa_j \sqrt{F_j F_{ref,j}}$; integrate it from one and use $\|u - v\|^2 = 2 - 2\,\mathrm{Re}\langle u, v \rangle$. This is an energy-form estimate, not an assumed Hilbert-space action of the singular interaction on arbitrary dressed vectors.

At a common endpoint $x_0(t) = x_1(t)$, the translation factors cancel from the relative overlap. If r_j encloses the square root of Eq. (210), then

$$\begin{aligned} \Gamma_{12} &= \mathrm{Tr}(\rho U_1^* U_0), \\ |\Gamma_{12} - \mathrm{Tr}(\rho e^{-iH_0(\eta_0 - \eta_1)})| &\le r_0 + r_1. \end{aligned} \tag{211}$$

Purification extends the statement to a mixed input with the same energy bound. A noncoincident final position supplies the relative translation $T_{x_1}^* T_{x_0}$ and belongs to an unrecombined experiment. The proper-age expression is exact when the difference form annihilates the relevant evolution; in general its error is now quantified. The actual source-lane clock equation also follows directly:

$$\begin{aligned} \dot{z}_j &= n_j [-i\Delta z_j + 2g w_j], \\ -\frac{d}{dt} \arg z_j &= n_j [\Delta - 2g\, \mathrm{Im}(w_j / z_j)]. \end{aligned}$$

Here $z_j \ne 0$, and w_j is the correlator in the actual common field state. The momentum term commutes with the source clock operator, but changes the field evolution and this correlator. An old isolated clock-rate certificate is therefore not assigned to these new trajectories.

18.3 Stability under additional source-profile and control data

The defined worldtube law has fixed lab-frame quadrupoles. A further physical profile, boost, control or microscopic-source assignment can be compared quantitatively rather than

identified with that law by notation. Suppose two such actual/reference propagators exist on the common energy domain. In their common frame let the difference form be $\delta h(t) - \delta v(t) \cdot P_f - \varepsilon \sum_a J_a \phi(\delta u_a(t))$, with bounded source $\delta h, J_a$ and $m_a(t) = \|\delta u_a(t)/\sqrt{\omega}\|^2 < \infty$. Any additional lapse-current difference can be included in δu_a and δh, retaining its even/counterterm terms. The annihilation form estimate between normalized vectors gives

$$|\delta q[\xi,\zeta]| \le \|\delta h\| + |\delta v|\sqrt{F_\xi F_\zeta} + \varepsilon \sum_a \|J_a\| \sqrt{m_a}(\sqrt{F_\xi} + \sqrt{F_\zeta}).$$

Consequently the squared vector distance is at most twice the integral of this expression along the actual/reference pair, by the same weak Duhamel argument. Finite energy ceilings close that estimate. This supplies an explicit extra error radius for independently supplied profile/control data without requiring an unweighted ultraviolet norm or fitting clock rates. Its work correction must likewise retain the corresponding control variation and energy balance. The numerical experiment below is for the exactly stated receiving action; no numerical profile or microscopic-actuator discrepancy is inferred for it.

18.4 Actual field response and common-arm coherence

At a frequency cutoff, the laboratory Heisenberg equation can be integrated exactly:

$$\begin{aligned} a_\sigma(k,t) &= e^{-i\omega t} a_\sigma(k,0) \\ &\quad + i\varepsilon \sum_{j,a} \Pi_j \int_0^t ds\, n_j(s) e^{-i\omega(t-s)} \\ &\quad \times u_{a,x_j(s),\sigma}(k) J_a(s). \end{aligned} \tag{212}$$

Smearing against an energy-admissible receiving test extends this identity as a form limit. The $J_a(s)$ are the actual interacting operators; their source and field backreaction has not been replaced by a classical force. Both arms use the same initial field operators. In the route-block state the cross block is $U_0 \rho U_1^*$, not a product of two bath states. Its trace is the interference operand in Eq. (211). The causal response is retarded in time. A spatially distant port additionally needs its actual local gauge-invariant observable and propagation geometry; a global frequency filter alone is not a local arrival channel.

18.5 Finite mechanical work and the phase reference

Let $E_{lab,j} = \langle H_j^{lab} \rangle$. The moving-frame identity is $E_{lab,j} = \langle K_j \rangle + v_j \cdot \langle P_f \rangle$. For a closed velocity cycle its net work is therefore

$$W_{lab,j} = \int dt \, \left[-\dot{v}_j \cdot \langle P_f \rangle + \dot{n}_j \langle H_0 - H_f \rangle \right].$$

This definition follows from endpoint energy balance and the common-form energy identity; it does not differentiate the ultraviolet-singular laboratory field at a moving point. A finite absolute payment ceiling is

$$W^{up}_{route,j} \le \mathrm{Var}(v_j) F^{up} + \mathrm{Var}(n_j)[\tfrac{1}{4} F^{up} + \varepsilon|e| + \varepsilon + 5c].$$

The even energy and counterterm remain in that expression. For nonzero endpoint velocities include $[v_j \cdot \langle P_f \rangle]_0^t$. An extra prescribed carrier rest energy m_{ext}, excluded from H_0, contributes its own point action and, for the sinusoidal closed speed cycle below, kinetic-energy variation $4m_{ext}[(1-v_*^2)^{-1/2} - 1]$. Beam splitters, traps and their calibration remain prescribed devices with separately supplied resources; the calculation is a driven receiving model, not a microscopic autonomous centre-of-mass theory. Their phases are included once in the total interference amplitude, as in Section 7.3.

19. A complete coherent local detector and record join

The two-arm evolution just derived is an actual source–continuum construction for the stated receiving action. It does not assume that every arm pair produces the quarter-cycle age separation used in the decoder calibration. We give a finite, explicitly paid route-label encoding that supplies a complete worked experiment, while retaining the unforced proper-age comparison as a distinct prediction.

19.1 Closed physical paths and a controlled source encoding

Take mirror paths with

$$\begin{aligned} v_{0,1}(t) &= \pm v_* \sin(2\pi t/T)\widehat{e}, \\ x_{0,1}(t) &= \pm \frac{v_* T}{2\pi}[1 - \cos(2\pi t/T)]\widehat{e}, \\ v_* &= 1/64, \\ T &= 2^{-20}. \end{aligned}$$

Both start and return to the same centre at rest. Their proper ages are equal. They still couple at different positions to one common field. Their total variation gives $B(T) \le 8v_* + 4v_*^2 = 129/1024 < 1$, hence $e^{B(T)} \le (1 - 129/1024)^{-1} = 1024/895$, and $\kappa \le v_* + v_*^2 = 65/4096$. The actual continuum route bound is

$$r^2_{route} \le 4T \frac{65}{4096} \frac{1024}{895}(C + \varepsilon + 8c). \tag{213}$$

For an initially coherent route state and one common prepared marker, the controlled actual-to-reference vector error is at most the largest branch radius; it is not the sum of two preparation errors. With the encoder absent and ideal unrecorded routing controls, equal proper ages give the concrete continuum consequence $|\Gamma_{unencoded} - 1| \le 2r_{route}$. The source and shared field thus supply a near-unit-overlap control for this short closed excursion. This is distinct from the encoded record experiment below, whose route correlation is paid by a different physical operation.

Prepare the original marker near age $T_0/4$, with its retained preparation-distance budget. After the closed excursion apply the positive finite route-controlled source pulse

$$\begin{aligned} B_{enc} &= \Pi_1 \otimes (I_{16} - S)/2, \\ B_{enc}^2 &= B_{enc}, \\ \int \lambda(t) dt &= \pi, \\ U_{enc}^{ideal} &= \Pi_0 \otimes I + \Pi_1 \otimes S. \end{aligned} \tag{214}$$

It acts on all 32 route/source states. It is a quantum controlled operation: the apparatus does not receive a classical copy of the unknown route label. Swap anticommutes with the source-lane Y, so the two reference outputs have opposite $-Y$ expectations and are decoded by the same complete source16 POVM. Swap commutes with H_f and preserves every source lane. The retained first-energy pulse lemma therefore gives

$$r_{enc} \le (1+2\pi)\sqrt{2C\tau_{enc}},$$
$$\tau_{enc} = 2^{-30}.$$

The pulse comparison factors out the same lane scalar phase on both route branches, exactly as in the retained finite-energy lemma. That scalar remains in the physical action and work account. The physical generator during the pulse is $H_0 + \lambda B_{enc}$, with the original even phase retained. Its peak is π/τ_{enc} and absolute switch-work payment is at most $2\pi/\tau_{enc}$. Those large finite controls are recorded explicitly. This encodes a route label; it is not an intrinsic gravitational age signal. In particular the mirror arms have equal proper ages, not the two distinct ages of the earlier calibration example. Without the encoder their overlap is given by the shared-field dynamics alone.

The small reference age increment obeys $\Omega T \le \Delta_{up} T < 0.01$ in every original lane. Thus the same declared phase allowance already present in the parent source decoder covers this reference timing. After ideal encoding the source field-energy ceiling is unchanged, since S commutes with H_f. The actual finite source decoder then obeys its original uniform form-energy and work bounds on either encoded input. No absolute hard energy phase needs to be measured by the source-only quadrature.

19.2 Coherent composition and the noise/work distinction

At the common return location couple the source to the finite decoder bit B, the actual nonuniform Chapter IV pointer, retained memory R and local access bit A. These are the coherent unitary/pulse dilations of the preceding integrated constructions. They act on marker and apparatus, with identity on the route register. The complete source, common field, bit/field/bath correlations and every residual pointer cloud remain.

The scalar response is the actual compact cap background plus the bit-controlled signed displacement. The latter is a transducer actuator, not an assignment of an uncomputed intrinsic remote cap mean. A local station at the common endpoint needs no remote bit-delivery assumption. For a remote station the physical bit or source signal must arrive through a specified causal channel before this interaction; no remote record is being inferred from the local construction.

Use a whole-pipeline hybrid: compare the actual routed and encoded joint input with the proper-age reference followed by ideal swap encoding; then apply exactly the same full detector/storage channel to both. Trace-distance contraction pays $r_{route} + r_{enc}$ once at the final readout. It is not used to transfer an unbounded energy moment. The parent detector moment is recalculated on its reference input using the unchanged field-energy ceiling and actual decoder work. The actual encoder energy has its separate, possibly much larger work ceiling. Both payments remain in the physical energy ledger. A small state error never licenses assigning the actual encoded state the reference energy. There is a separate actual-energy envelope. At the closed return, $\langle L \rangle \le E_* e^{B(T)}$, and bounded-pulse energy balance through encoding and decoding gives

$$F_{actual,in} \le 2\left[E_* e^{B(T)} + W^{abs}_{enc} + W^{abs}_{dec} + c\right]. \tag{215}$$

The physical field-gate resource ledger uses this larger ceiling, not the reference noise moment used in the final probability comparison. For its actual energy-admissible cap test f_t, put $\mu_t = \|f_t/\sqrt{\omega}\|^2$ and $\dot{\mu}_t^{test} = \|\dot{f}_t/\sqrt{\omega}\|^2$. The conserved pointer momentum second moment p_2 and annihilation form estimate bound the controlled cap switching work by

$$W^{abs}_{cap} \le 2\sqrt{p_2}\int dt\,[|\dot{h}|\sqrt{\mu_t} + |h|\sqrt{\dot{\mu}_t^{test}}]\sqrt{F_{actual}(t)} + p_2\,\mathrm{TV}(s).$$

This follows from $|\langle P\phi(f)\rangle| \le 2\sqrt{p_2 \mu_f F_{actual}}$ on the joint state, with no bit/field independence or fourth moment. For the stabilized gate's common-form graph, let $a(t)G \le q_t + b(t)$ and $|\dot{q}_t + \dot{b}(t)| \le \ell(t)G$. Its actual energy satisfies $\mathscr{E}(t) \le \mathscr{E}(0)\exp\int \ell/a$, using Eq. (215) at entry. The retained smooth compact test/gate and positive stabilizer give finite graph coefficients and cap-test norms. This is a finite functional ceiling for actual gate work even if the actual energy greatly exceeds its reference value. Global background vacuum-ramp and material actuator work remain separately specified external payments, as in the parent; they are not set to zero or inferred from the noise ceiling.

Consequently the worst-label final ready error obeys

$$e^r_{new} \le e^r_{record} + r_{route} + r_{enc},$$
$$e_{all} \le f + (1-f)e^r_{new}. \tag{216}$$

Here the parent record bound already includes preparation, source decode, pointer loading, finite writing, initialization, hold, aperture and local read error. Those terms are not charged a second time. The native operand used here is the parent's conservative `full_error` return; its smaller refined aperture/read display is not silently used to improve this budget. An encoder area-calibration error a_{cal} adds at most $|a_{cal}|$ by bounded-generator Duhamel. The numerical example specifies nominal encoder area exactly, just as its nominal trajectory; it assigns no achieved control calibration. The native six-lane evaluation certifies $e^r_{new} < 0.0016$ with the displayed chosen controls; f remains the supplied per-label failure ceiling. The particular $f = 10^{-4}$ line is an explicit budget, not an achieved failure measurement.

For the same coherent ensemble, route visibility and this readable route record obey the integrated tight theorem

$$V \le 4\sqrt{pq\,e_{all}(1-e_{all})}.$$

The marker-only decoder, writing and access operations leave the unsorted route marginal exactly unchanged. They make an existing route correlation readable; they do not additionally suppress that marginal merely because a later observer reads it. Conditional erasure or route feed-forward is a separately specified experiment.

19.3 All outcomes, resources and latency

Retain read results 0 and 1, every preparation/aperture/control failure flag, and all unobserved recipients in one trace-preserving instrument. A ready-branch conditional calculation never discards its failed input weight from the all-attempt ledger. The source decoder and record pulses remain active finite-duration operations with their original positive completions and pointer kinetic energy. The new experiment adds the route work ceiling, encoder capital/peak/switch work and durations $T+\tau_{enc}$ to those parent resources. The long analog and logical holds remain; the source clock certificate applies at source entry, not through the entire long receiving latency. Microscopic actuator, absolute length/time and mass calibration remain physical model assignments.

20. Ordered physical histories and repeated access

These extensions follow the route and coherent-record join. An ordered word is assigned a unitary only after its physical event/route/control incidences have been specified. For such unitaries on the one full marker space define

$$G_{wv} = \mathrm{Tr}(\rho U_v^* U_w),$$
$$(\rho_{route})_{wv} = a_w a_v^* G_{wv}.$$

The matrix G is positive semidefinite with diagonal one: purification expresses it as the overlap matrix of normalized conditional marker vectors (with the indicated index convention). Thus the route density operator is normalized and positive for normalized route amplitudes. Every multi-history interference measurement follows from that same operator and the stated analyser. On a declared accessible domain the conditional states are $\mathscr{A}(U_w \rho U_w^*)$; equality of these states means identical probabilities for its marker-only POVMs. It does not imply equality of the coherent cross blocks, nor does a computational history label alone supply an extra observable. Equal signed action and final address therefore leave a concrete question about the ordered physical propagators rather than an unassigned rule creating additional states. In the moving-source model, $[H_0, P_f]$ need not vanish as a formal core commutator, so ordered velocity controls need not be determined by their integrals. The actual form propagators, not that potentially unbounded commutator, define the history comparison.

20.1 A finite repeated-read theorem with correlated noise

Let Q0,Q1 be the orthogonal populations of the retained memory. An ideal copy-read isometry has $W|j\rangle_R|0\rangle_A = |j\rangle_R|j\rangle_A$, extended unitarily on its complete finite space. A finite positive controlled flip realizes this operation with area π, peak π/τ and absolute switch-work at most $2\pi/\tau$; the memory/bath drift must be charged as in the finite pulse construction. Diagonal interactions conserve Q populations even with a correlated bath. For transverse disturbance $V_\perp$, retain each interval's trace-distance exposure $d_k \le \int \|V_\perp(t)\| dt$. Suppose actual read k differs from the ideal nondestructive read by δ_k on the complete reachable joint state, including all earlier read registers and the bath. Successive hybrid replacement and trace-distance contraction give

$$\Pr\left\{\begin{array}{c}\text{any wrong stored/read label}\\ \text{through read } N\end{array}\right\}$$
$$\le \min\{1, e_{write} + \sum_{k=1}^{N}(d_k+\delta_k)\}.$$

No statistical independence is used. For worst-label uniform bounds $d_k+\delta_k \le b$, an allowed total error e_{max} permits $Nb \le e_{max} - e_{write}$; this is a finite guaranteed-read budget, not a derived thermal decay rate. Actual sequential joint-state bounds are essential: separate one-time marginal read errors do not establish this premise. Classical repetition improves reliability exponentially only under its additional conditional-independence assumptions.

Every read needs a recipient. An occupied register cannot be reused as a blank without an explicit reversible transfer or disposal/reset channel. Exact discrimination of N complete histories requires N orthogonal recipient states; the integrated approximate label-capacity statement gives its finite-error version. Resetting the local cell does not erase its globally retained source/bath/history recipients. A thermal heat claim additionally needs a specified reservoir, initial correlations and cyclic protocol. No universal heat-per-Write number is assigned here.

21. New source-bound receiving values and evidence

All displays round upward; exact native rationals remain in the replay. The all-attempt column uses the explicitly declared budget $f = 10^{-4}$. The additional encoder peak is below 3.374×10^9 and its absolute switch-work ceiling below 6.748×10^9 in original model units. Route mechanical work obeys the full source-dependent bound in Section 18; trajectory carrier mass and splitter/trap calibration are not assigned numerical physical values.

The new native work uses the installed public DomainSession route, SLC-GEN5-R1 / SLC-GEN5-CEV1-R1 and Starbreaker SB-GEN3-ACCUMULATION-R1. One exact six-lane bound calculation and one full-source32 encoder certificate complete. A first 96-column certificate exceeded the installed 64-column word limit; its rejection is retained. A corrected 32-column sum of residual Grams certifies the same three identities, and the completed first bound call was adopted rather than repeated. A subsequent binding-index preparation error started no native work. A native checkpoint and both closed sessions remain. No Duck campaign or installation changed.

The continuum propagation, energy and coherent-join statements are analytical theorems, with independent arranged arithmetic and assigned finite mechanism comparators. Finite field fixtures do not replace the continuum, its ultraviolet domain, or the actual preparation. Offline replay authenticates the used inputs, receipts and exact matrices; it performs no native science. Existing source and supplement results are retained unchanged. The original detector export ZIP was unavailable at its recorded TEMP path; its retained source members are authenticated against the completed manifest, not described as the old ZIP.

Lane	Route radius	Encoder radius	Ready error	All-attempt error
01	6.242E-7	6.012E-7	0.0015814301	0.0016812720
02	6.918E-7	6.448E-7	0.0015815514	0.0016813932
03	0.0000042811	0.0000033658	0.0015898752	0.0016897162
12	6.275E-7	6.032E-7	0.0015814359	0.0016812777
13	0.0000041185	0.0000032394	0.0015894322	0.0016892732
23	0.0000039443	0.0000031040	0.0015889641	0.0016888052

The exact decoder archive was recovered from the detector's retained inputs.

Completed and remaining physical objects. For the specified moving rigid-source action we have full source/shared-field dynamics, a proper-age comparison, actual retarded source response and a coherent co-located detector/record construction. The finite paid encoder is a supplied receiving operation. It is not an independently established primitive route-selection law, a positive intrinsic remote cap mean, or an achieved material calibration. A remote intrinsic-field protocol must specify its actual local observable, spatial propagation, extraction and signal/noise law; a local encoded bit is not substituted for that port. An autonomous quantum centre-of-mass and microscopic control mechanism are also beyond the prescribed-worldtube action. These are exact physical joins to carry forward, alongside every retained counterterm and recipient.

Reproduction. Run `python3 checks/verify.py` from this folder for the independent arithmetic/custody and finite mechanism checks. Build with `python3 build.py` using the retained existing Tectonic executable and cached packages. No network, compiler install, Duck cycle, cloud, Git publication or I9 operation is required.

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